

Advanced Materials - PCB & CCL

#AI

Companies: ISU Petasys, TTM Technologies, Elite Material

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STOCK INFORMATION

US\$ bn	ISU Peta	TTM	EMC
M.cap	6.6	13.9	58.2
Rating	Buy	Buy	Buy
TP	₩270k	\$236	TWD10k
Upside	123%	79%	93%

KEY VALUATION MATRIX (2 Oct '26)

FYE Dec	ISU Peta	TTM	EMC
PER (x)			
FY26E	32.2	27.5	40.3
FY27E	16.6	17.6	19.7
FY28E	11.2	13.2	12.8
EV/EBITDA (x)			
FY26E	22.6	20.0	29.7
FY27E	12.1	12.8	14.8
FY28E	8.1	9.6	9.5
EPS Growth (%)			
FY26E	65	95	215
FY27E	94	57	104
FY28E	48	33	54

Goldilocks

We resume coverage of the PCB/CCL industry, focusing primarily on Google's TPU supply chain, which we believe will deliver the strongest and most sustainable growth through 2028 and beyond. We forecast aggregate TPU volumes to more than triple between 2025/26 and 2028, far outpacing competing platforms. Critically, we project a surge in the content value of next-generation TPU ASICs. As Google integrates more advanced logic and HBM chiplets into a single compute module, we project TPU content value to increase five- to tenfold over the next three to four years. Accordingly, we estimate that the addressable PCB content value alone could quadruple from 2026 to 2029. Equally vital is Google's longstanding leadership in optical connectivity for its Superpods. The upcoming doubling of switch speeds and interconnect bandwidth will act as a secondary tailwind, further lifting PCB content value. Consequently, we expect the PCB/CCL market to experience continuous pricing power and a possibility of LTAs forming, mirroring the supply dynamics seen in the memory industry. Within the key TPU vendor matrix, we resume coverage of Isu Petasys and EMC with BUY ratings and target prices of KRW 270k and TWD 10,000, respectively. We also initiate coverage of TTM with a BUY rating and a target price of \$236. Risks include end demand, execution, and US policy on AI data centre expansion.

Why Google's TPUs?

We forecast TPU's volume to more than triple to 12m units in 2028 (8.5mn in 2027) from 3.6mn in 2026, encompassing both in-house deployment and allocations to third-party AI labs via ASIC services. The strong volume is evident by its massive \$520bn cloud service backlog orders in C2Q26. At the same time, TPU is expected to quadruple in TPU V9 (mid-28) vs V8 (2H26) and could double again in V10 (2029) as Google squeezes more chiplets into its ASIC designs. The complexity of design for PCB (upgrade to 50L+ or HDI) and CCL (upgrade to M8/M9 material or even PTFE) represents at least 50%-100% expansion in ASP in each generation. Furthermore, networking is expected to see a similar upgrade trend in PCB/CCL, as scale-up architectures and bandwidth requirement rise, especially from 400Gbps to 800G/1.6Tbps.

LTA likely becomes new norm under strong demand and supply constraint

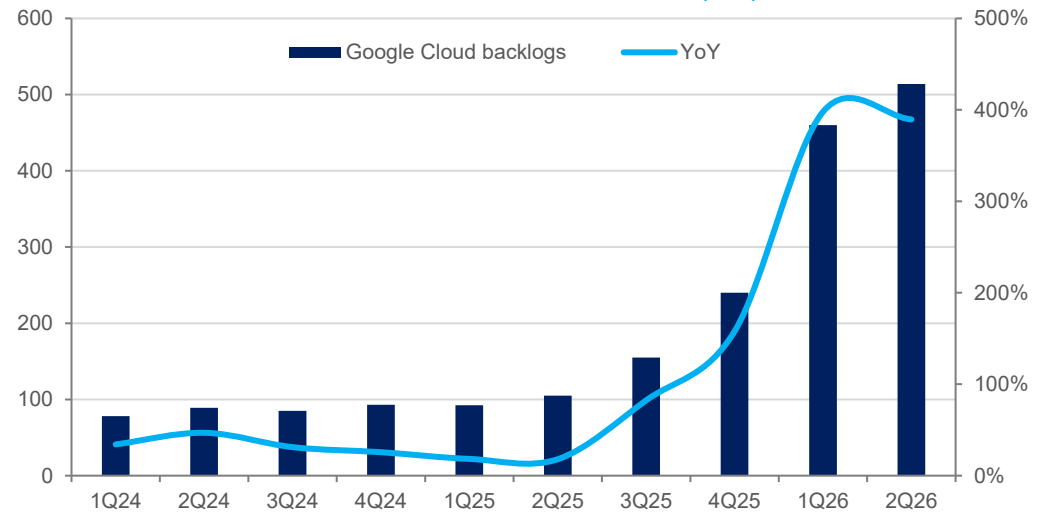
Our supply chain survey indicated increasing discussion about long-term agreements (LTA) over the PCB and CCL supply chain with CSPs. Terms may be different from vendor to vendor, and most of them are still under discussion at this stage. However, those discussions send a clear message that end users are securing high-end PCB/CCL capacity by prepayment and LTAs for the next few years, because the overall industry may continue to remain undersupplied. Also, we reckon prepayment 1) will ease supply chain pressure on capex/cash flow, 2) make the supply chain more willing to invest given longer visibility and secured shipment, and 3) improve supply chain management with better profitability. We believe this will further boost P/E for the supply chain, if LTAs are confirmed, with longer visibility and stable order backlogs.

We prefer TPU supply chain into 2027/28: buy on EMC, ISU and TTM

The TPU supply chain will be the biggest winner into 2027/28 with strong unit growth and content value upgrades. Shipments are expected to reach 8.5/12m in 2027/28 vs 3.6m in 2026. Content value per rack for PCBs is up to USD30k-40k and CCL USD12k-18k for TPU v8 (vs USD20k/7k in v7). We expect another 40%-50% in v9 with HDI adoption. We like EMC for its leading position and expect TPU/800Gbps switches to become the company's key earnings driver. We set our TP at NT\$10,000, with 30x FY27/28 PER. ISU is uniquely positioned to capture these secular trends due to its high exposure to Google and networking. We have a BUY and a TP of KRW270,000 on 30x FY27/28 PER. Lastly, TTM is the best proxy in the United States to capture both AI server and national defence trends with better profitability and irreplaceability

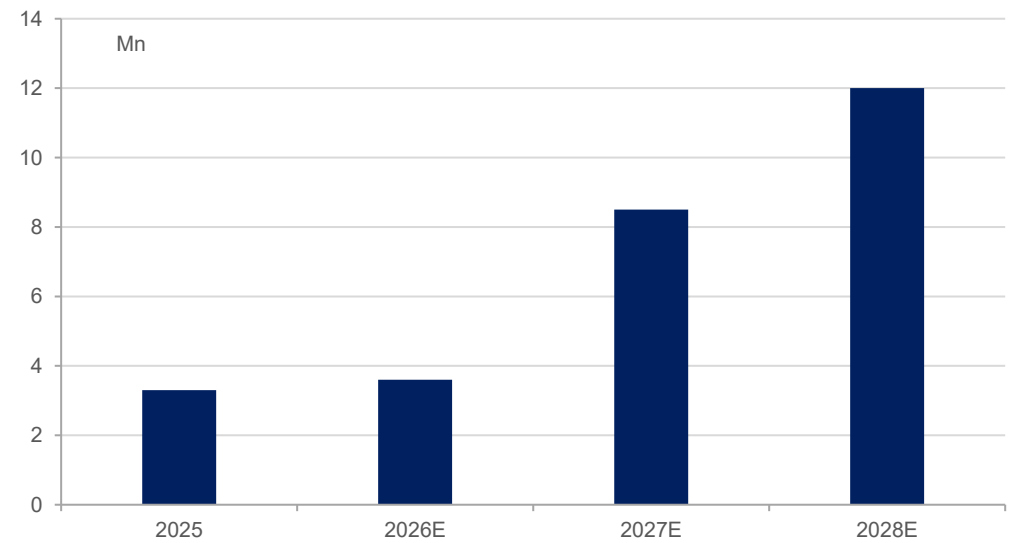
What matters to Google's TPU franchise

FIGURE 1: BACKLOG ORDERS OF GOOGLE COMPUTE CLOUD SERVICES (\$ bn)



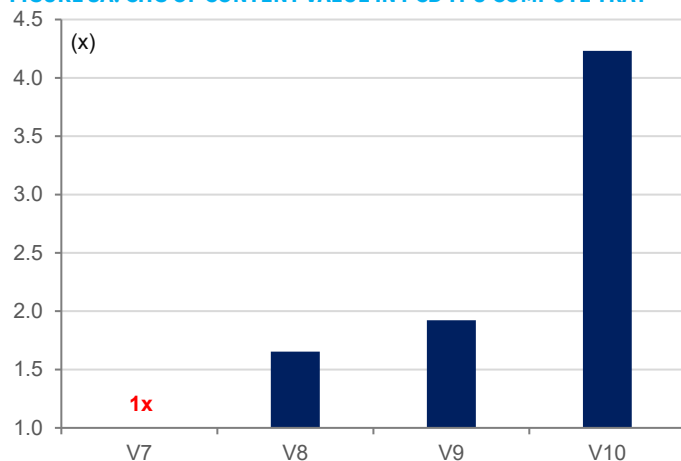
Source: Google & Aletheia/TAG (Sep 2026)

FIGURE 2: ESTIMATED TPU VOLUME



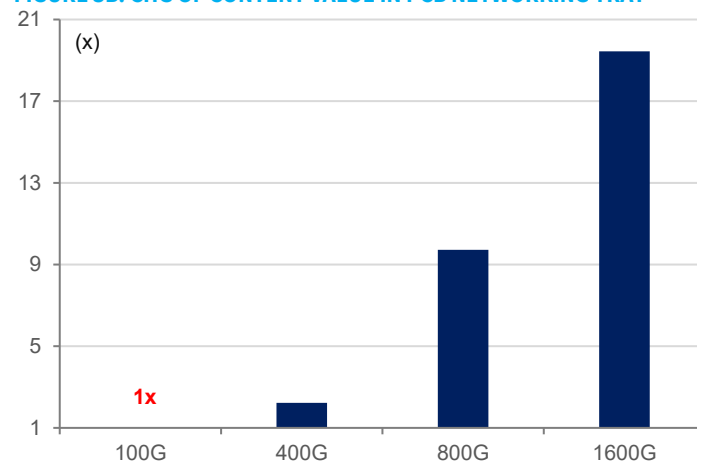
Source: Google & Aletheia/TAG (Sep 2026)

FIGURE 3A: CHG OF CONTENT VALUE IN PCB TPU COMPUTE TRAY



Source: Aletheia/TAG

FIGURE 3B: CHG OF CONTENT VALUE IN PCB NETWORKING TRAY



Source: Aletheia/TAG

GPU/ASIC volume, content value and price hikes drive PCB/CCL to another strong cycle in 2027/28

We are positive on the PCB/CCL sector into 2027/28, with a strong upgrade cycle, and the whole industry will remain undersupplied because our checks indicate 1) continued AI device unit growth, supported by increasing CSP capex, 2) PCB design complexity will boost content value (increasing processing time and lowering yield rate), 3) CCL material upgrade to support higher data transmission and power supply, and 4) price hikes in the near-term because of the supply shortage and LTAs in the longer-term for a better demand outlook.

What is not in the price yet?

The sector has outperformed year-to-date, but we believe the PCB/CCL upgrade cycle will continue to surprise the market because 1) stronger AI device unit growth (especially Google TPU, of which we project 8.5/12m units in 2027/28), 2) significant content value growth via PCB design complexity (higher layer counts, advanced form factor like HDI/mSAP, longer processing time) and CCL material upgrades (M8/M9, PTFE), 3) wider AI end devices (GPU, ASIC, switch, CPU and LPU, for example) will all require high end PCBs to support performance and 4) better cash flow and profitability via prepayment and LTAs.

The overall industry is under a severe supply shortage at this stage, especially high-end products, with the key bottleneck being in upstream material (CCL, glass fiber and copper foil) as well as PCB equipment, based on our supply chain checks. The longest lead-time for certain PCB equipment (like drilling machines) can be up to 36 months, suggesting strong surge in demand from the supply chain. We don't see any sign of the tightness easing in the near term.

AI server unit growth and spec upgrade in 2027/28

We estimate GPU & ASIC volume will grow approximately 50% in CY2027 and CY2028, reaching a potential 34mn units in CY28, up from 15mn units in CY26. The key driver is the expected unit surge in both CSPs' ASICs as well as AI labs, that is, OpenAI and Anthropic. Their aggregate volume is forecast to rise from 5.5mn -6mn in CY26 to reach 12mn -13mn units in CY26 and further increase to 20mn in CY28, implying a quadrupling in a three-year period. Some of the notable drivers including the following:

- OpenAI's direct procurement compute power (excluding CPS and neoclouds) is expected to reach 11Gw in CY28, up from 3.3GW in CY27 and a mere 1GB in CY26. This includes AVGO's-sponsored ASIC of 1.3GW (Jalapeno) and 5GW in CY27/28 respectively, while the remaining goes to AMD (8GW in the three-year period) and, to a lesser extent, NVDA;
- Similarly, Anthropic's own compute capacity is likely to reach 13GW in CY28, up sharply from CY27's 6GW. The majority of this comes from AVGO (TPU V8i and V8.5) while 5GW is supplied by AMD of in two years;
- Meta is likely to build 3GW of its own compute on its ASIC (via AVGO) in CY27/28 using its MTIA400/450 ASIC. We note Meta's previous ASIC program in 2025/26 has been scaled back and pushed out so this marks the second attempt at ramping up its own silicon solution.
- We estimate Google's own compute TPU volume is expected to increase to 5.5mn units in CY28 from 4mn -4.5mn in CY27 and ~3.5mn in both CY25/26. While the unit growth appears to be modest relative to the AI labs, we note the compute capacity is likely to surge given the TDP for V8 and V9 is materially higher than those of V6/7.

While NVDA's GPU volume is likely to moderate and market share is expected to decline (as most of its customers are building their own ASIC and diversifying to other suppliers), it will remain the dominant supplier to most CSPs, neoclouds and enterprises. Moreover, NVDA, along with industry is migrating to a multi-rack or large pod structure to offer aggregated

system compute and memory. NVDA's ability to offer comprehensive GPU, CPU, LPU, storage and networking racks sustain its competitive edge in the fast-moving environment.

FIGURE 4: ESTIMATED GPU & ASIC UNIT FORECASTS (MN UNITS)

	2025	2026E	2027E	2028E	Remarks
NVDA	5.4	7.6	9.5	11.4	There are ~1mn units of GB300 in production in CY27. The mix in CY2028 is pending on the dual- or quad-GPU Rubin Ultra and the design status of Feynman.
-Rubin	-	2.0	6.5	TBD	
-Rubin Ultra	-	-	2.0	TBD	
AMD	1.9	1.6	2.0	3.0	OAI and Anthropic recently upped orders for CY2028
Google TPU	3.3	3.6	8.5	12.0 ¹	TPU units include Anthropic's volume via AVGO. For CY27, we expect an equal split between V8i&t variant in 2027 and limited volume of V8.5 (Whalefish). V9 HVM expected in mid-2028.
-V8i+t	-	1.0	8.2	10.5	
-V9	-	-	-	1.5	
Trainium	1.9	1.6	2.0	2.0	Status of T4 TBD
OpenAI	-	0.2	1.0	3.5	1.3/5GW deployment in CY27/28
MTIA	0.2	0.2	0.7	2.0	MTIA450 more meaningful volume in 2027/28
Others	0.2	0.2	0.3	0.5	Volume of Maia300/400 to be more meaningful
Total	12.9	15.0	24.0	34.4	Expect ~50%CAGR CY26E-28E

Note: 1-We assume each V8.5 (Whalefish) is eqvl of 2x V8i (Sunfish). 10GB of deployment in CY28 is 6.5mn units of V8i or 3.3mn units of V8.5 (Whalefish).
Source: Aletheia/TAG estimates

AI server performance to jump significantly every generation

As shown in the table below, we see huge upgrades in compute power and TDP every generation. PCBs and CCL will upgrade along with AI server specs to reduce data loss and improve signal performance. PCB design is getting more complex from 20L -30L to 50L+ in the future or the structure will likely be migrated from HLT to HDI. For CCL, we witness a clear trend of upgrade material from M7 to M8 now and will see more adoption of the M8.5/M9 going forward, boosting the ASP (a doubling from M7 to M8; M8 to M9 also doubling).

In the future, along with strong agentic AI demand, we believe AI server design will further advance exponentially in the next two to three years. TPU's TDP is expected to grow 10x from v8 to v10 within two to three years to further drive-up PCB/CCL's content value significantly. AMD will equip PCBs with 50L+ into the MI500 generation and Open AI's aggressive hardware production pipeline will provide upside surprise to the market. We can expect PCB/CCL to upgrade accordingly. We will have a more comprehensive analysis in the next section.

FIGURE 5A: SPECIFICATIONS OF SELECTED GPU & ASIC DEVICES

	Rubin	Rubin Ultra ¹ (Dual-die)	MI500 ³ (Quad-die)	TPU V8i (Sunfish)	TPU V9i (Humufish)	OpenAI (Jalapeno)
Logic	2xN3	2xN3	4xN2+2xN3	2xN3P	4xN2P+4xN3P	1xN3P
I/O	1xN3+1xN4	1xN3+1xN4	4xN3	1xN3E	4xN3P	1xN3E
HBM	288GB/4	192GB/4 ² 256GB/4E	512GB/4E ³	288GB/3E	384GB/4E	216GB/4
Bandwidth	21TB/s	21/32TB/s	Up to 66TB/s	10-15TB/s	Up to 49TB/s	15.4TB/s
TDP	2.3kW	2.6kW	4.5-5kW	600W	4-5kW	700W
FP4	50PFLOPs	<50PFLOPs	~100PFLOPs	10.1PFLOPs	N.A.	13.4PFLOPs
HVM	2H26	2H27	4Q27	4Q26	Mid-28	4Q26

Note: 1-There is a quad-die Rubin Ultra SKU. 2-The final specs of Rubin Ultra are yet to be determined. At the time of writing, there are three options consisting of 192GB of HBM4 (8-Hi), 256GB HBM4E (8-Hi) and 386GB HBM4E (12-Hi).

3-There is a dual-die (with 12-Hi) MI500 GPU likely to go to the Helios rack. The quad-die GPU is equipped with 12-Hi HBM4E.
Source: Aletheia/TAG

FIGURE 5B: SPECIFICATIONS OF SELECTED MULTI-RACKS OR PODS

	Rubin	Rubin Ultra ² (Dual-die)	MI500 ³ (Quad-die)	TPU V8i (Sunfish)	TPU V9i (Humufish)	OpenAI (Jalapeno)
Scalability	576 ¹	576 ¹	512	1,152	1,600 ⁴	2,048
Networking	Spectrum-X	Spectrum-X	T100 or S1 G300	ICI	ICI	TH6
System compute	29EFLOPs	<29EFLOPs	51EFLOPs	12EFLOPs	N.A.	27EFLOPs
System memory	166TB ⁵ , 12PB/s	111-147TB ⁵ , ~20PB/s	262TB, 34PB/s	332TB, 17PB/s	614TB, 78PB/s	432TB, 32PB/s
Device/tray Tray/rack	4/18	4/18	1/16	4/8	N.A.	8/16
Rack/pod	8	8	32	36	N.A.	16
TDP/rack	225kW	260kW	~100kW	<80kW	N.A.	~125kW
Rack HVM	1H27	1H28	1H28	Mid-27	2H28	2H27

Note: 1- We assume 8 racks as a pod, but NVL Super pod can support as many as 16 racks or 1,152 GPUs. 2-There is a quad-die RU SKU. 3-The final specs of AMD's super pod structure are yet to be finalized. There is a dual-die MI500 GPU (each comes with 8 stacks of 12-Hi HBM4E) likely to go to the Helios rack. 4-Estimate based on TPU V8i. 5-VR200 rack comes with 576TB of high-performance eSSD or 4,608TB of eSSD with eight racks pod. The eSSD density for Rubin Ultra is expected to rise further.
Source: Aletheia/TAG

Summary: PCB/CCL spec upgrade and key suppliers in AI solutions

The table below shows the key players' solution and their PCB/CCL specifications as well as major suppliers. For **PCB**, we see an increasing layer count from 20L–30L now to 50L+ in the coming years and moving from HLC to HDI for more complex design. Our supply chain check suggests AMD's MI500 series as well as Nvidia's LPU PCB will likely adopt 50L+ design. This represents a longer processing time with a lower yield rate, and design complexity will support further ASP hikes. For **CCL**, M8 is becoming a standard material for AI servers/networking and we are going to see more advanced materials into 2027/28 (M9/PTFE). CCL is the most critical component in PCB and every generation upgrade (M7 to M8, or M8 to M9) implies the ASP will nearly double.

Nvidia: It started to adopt M8 material from the GB200 NVL72. Rubin is expected to bring another modest upgrade: 1) introducing mid-plane with 44L design and M8 (or M8.5) CCL material, 2) most of the rest PCB/CCL remain on M8 with higher layer counts. The company is evaluating PTFE material for its switch board in the next generation. Also, we think Nvidia's LPU can generate incremental demand for PCB/CCL with its 52L design with M9/M8 materials. PCB suppliers for LPU will be VGT, WUS and TTM mainly, while, in our view, CCL suppliers are Shengyi (major) and EMC.

Google: TPU is expected to see the biggest upside in the PCB/CCL addressable market in 2027/28 thanks to both unit growth and content value upgrades. The next section will have a comprehensive analysis.

Amazon: Trn 3's air-cooled solution will remain at a 26L PCB design (with M8 material), while the liquid-cooled version will migrate to 40L. for Trn3 is solely EMC and TUC are the only CCL suppliers for the Trn 3. We expect to see a more significant upgrade for PCB/CCL in the next generation Trn 4 (higher layer counts, upgrade to M9?)

AMD: AMD is another key growth driver for the sector, with meaningful volume into 2027/28. The MI450, with 48L/M8 design, will start to contribute from next year and we believe AMD's MI500 series will likely adopt a 50L+ PCB design with M9 materials. This will represent a huge

increase in content value. SCC is its major source for PCB supply while CCL suppliers are EMC and Doosan.

OpenAI: Our supply chain checked OpenAI's Jalapeno roadmap and its pipeline to reach 1.3GW and 5GW deployment in 2027/28 is aggressive. We have limited visibility of Jalapeno's hardware specs. However, given its strong performance (~700W TDP), we can assume it will be equipped with similar specs to TPU v8 for PCB/CCL (30-40L PCB with M8 materials).

FIGURE 6: PCB/CCL SPECS AND KEY SUPPLIERS FOR AI SERVERS

	Code name	Timeline	PCB specs.	PCB suppliers	CCL specs.	CCL suppliers
Nvidia	GB200	1H25	22L HDI 24L (switch)	VGT, Unimicon VGT, TTM, Unimicon (switch)	M8/M4 M8/M2 (switch)	Doosan Shengyi, EMC (switch)
	GB300	1H26	22L HDI 24L (switch)	VGT, Unimicon VGT, TTM, Unimicon (switch)	M8/M4 M8/M2 (switch) M8/M4	Doosan Shengyi, EMC (switch)
	Vera Rubin	2H26	26L HDI 32L (switch) 44L (mid-plane)	VGT, Unimicon, ZDT VGT, WUS, TTM, Unimicon (switch)	M8/M4 M8 (switch) M8 (mid-plane)	Doosan Shengyi, EMC, Nanya (switch)
	Rubin Ultra	2H27	30L HDI 32L (switch) 44L (mid-plane)	VGT, Unimicon, ZDT VGT, WUS, TTM, Unimicon, SCC (switch)	M8/M6 M8/M9+ PTFE (switch) M9 (mid-plane)	Doosan Shengyi, EMC/Taiflex (switch)
Google	TPU v7 Ironwood	1H26	26L	ISU, LCS, WUS, TTM	M7	Panasonic, EMC
	TPU v8i Sunfish	4Q26	32L M+M	ISU, LCS, WUS, TTM, Dynamic	M8/M6	Panasonic, EMC, Nanya
	TPU v8t Zebrafish	1Q27	34L N+M	ISU, LCS, WUS, TTM, GCE, Dynamic	M8/M6	Panasonic, EMC, Doosan
	TPU v9 Humufish	2028	HDI	ISU, LCS, WUS, TTM, Unimicon, Dynamic	M9/M8	Panasonic, EMC, Doosan
Amazon	Trainium 2e	1H26	26L	GCE, Shengyi, FHT	M8	EMC, TUC
	Trainium 3	2H26	26L/40L 22L (switch)	GCE, Shengyi, FHT, WUS, ZDT	M8	EMC, TUC
	Trainium 4	2H27	34L 30L (switch)	GCE, Shengyi, ZDT	M9?	EMC, TUC, Nanya
AMD	MI450	4Q26	48L	SCC (major), Unimicon, Dynamic	M8	EMC, Doosan
	MI500	4Q27	48-52L	SCC (major), Unimicon, Dynamic	M9	EMC, Doosan
OpenAI	Jalapeno	4Q26	40L	GCE and others	M8?	EMC, Doosan

Source: Aletheia/ TAG

FIGURE 7: NVIDIA LPU PCB/CCL SUPPLY CHAIN

	Code name	Timeline	PCB specs.	PCB supplier	CCL specs.	CCL suppliers
Nvidia	LPU	Late 2H26	52L	VGT, WUS, TTM	M9/M8	Shengyi, EMC

Source: Aletheia/TAG

Google's TPU to undergo rapid spec upgrades in next several years; its PCB/CCL will benefit accordingly

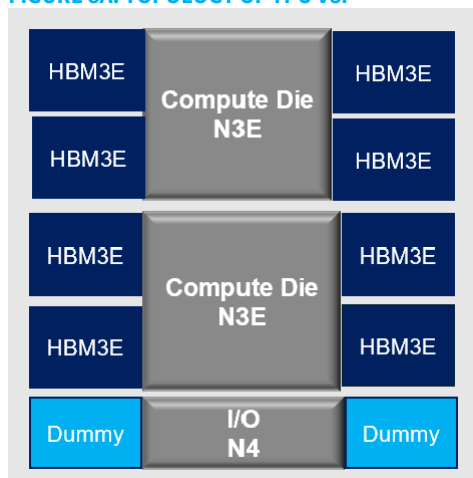
The figures below illustrate the rapid production transition and specification advancement of Google's TPU roadmap in the next several years. We highlight the following considerations:

► The current V8i variant, which is undergoing HVM (through 1H28), receives a modest upgrade vs TPU V7 in terms of design and specification. For the latter, there are two logic dies and four HBM stacks encompassing TDP of approximately 1,000w. In TPU V8i, the TPD rises by 30% with the addition of one logic tile and two HBM stacks.

► We expect the next generation of TPU V9 to undergo a massive uplift in design complexity, TPD and compute capability. We estimate there are 12 logic dies (4x times more than V8i) and 12 HBM stacks (double that of V8i) to be the first to use TSMC's N2 process node. Given the rapid rise in chiplet quantity, we estimate the reticle size of the design to be 9.5x reticle size. The compute capability is likely to surge accordingly with the TDP of both training and inference to reach 4,500w, more than triple compared to TPU V8i. We expect TPU V9 to begin HBM in mid-2028 and run through 2029.

► We think TPU V10 will experience the next wave of significant specification and performance uplift as the number of logic chiplets rises further to 30, 2.5x vs TPU V9 and they are all to be fabricated on TSMC's N2P geometry as well as being encompassed with 3D packaging for the first time. In addition, the number of HBM stacks will rise by one-third to 16 stacks. We estimate the reticle size is likely to increase to 15x, 50% larger than TPU V9 and critically the TDP to surge to 10,000w if not higher!. TPU V10 is expected to enter HBM in 2029.

FIGURE 8A: TOPOLOGY OF TPU V8i



Source: Aletheia Capital/TAG

FIGURE 8B: TOPOLOGY OF TPU V9 (HUMUFISH)



FIGURE 9: SPECIFICATIONS OF TPU'S ASIC

Specifications	TPU V8i 2H26/27	TPU V8.5 2H27/28	TPU V9 Mid-28	TPU V10 2029
Compute die	2x (N3)	4x (N3)	4x (N2P)	16x (N2P)
Core logic/base die	-	-	4x (N3P)	8x (N2P)
I/O die	1x (N4)	2x (N4)	4x (N3P)	6x (N2P)
HBM	8x (12-Hi HBM3E)	12x (8-Hi HBM3E/4)	12x (8-Hi HBM4E)	16x (8-Hi HBM4E)
Total dies	11x	22x	24x	46x
2.5D packaging	CoWoS	CoWoS	EMIB-T/penal	EMIB-T/panel
3D packaging	No	No	No	Yes
Reticle size	~2.5x	~5x	9.5x	15x
Estimated TDP	~1,300W	~2,600W	~4,500W	~10,000W

Source: Aletheia/TAG estimates

Implications for PCB/CCL

We think the TPU supply chain will enjoy the biggest growth in 2027/28 along with strong unit growth and content value upgrades among all the GPU/ASICs. Shipments are expected to reach 8.5m/12m in 2027/28 vs 3.6m in 2026. We view increasing compute power, higher HBM density, and a doubling of ICI bandwidth as pushing PCB designs toward significantly higher routing density, tighter L/S, more advanced CCL, smaller via structures, and multi-stage lamination to maintain signal integrity and shorten electrical paths.

We observed that TPU's PCB ASP has risen 2.5x from TPU V5 to V8, and 4x for the 400Gbps switch to 800Gbps. In addition, we estimate 2.6x growth in TPU unit demand from 2025E-27E and 3x expansion in the 400bps/800Gbps AI switch market by 2028 versus 2025, or 30% growth in total, making TAM growth exponential in these years.

The table below illustrates the structure of Google TPU v8. One rack will contain eight compute trays, with each tray equipped with four TPUs, four NICs, four optical modules as well as two CPUs (Axiom). TOR (Top-Of-Rack)/OCS will connect those racks together to form a superpod (via the Virgo network) and the Juniper network to connect with other functions. The increasing compute power and connectivity represents significant content value growth for PCB/CCL based on those performance upgrades and data transmission.

To support the performance upgrades, TPU PCB architecture is shifting from standard multilayer (Ghostfish) to N+M structures using dual-grade CCL and multi-lamination in Zebrafish and Sunfish, and ultimately to HDI PCB in Humufish. These transitions lift PCB ASPs by 30% in Ghostfish, 80%–90% in Sunfish/Zebrafish. **Based on our estimates, content value per rack for PCB is up to USD30k-40k and CCL USD12-18k for TPU v8 (vs USD20k/7k in v7). We expect another 40%–50% in Humufish with HDI adoption. Into TPU V10, given that TDP will further double to 10kW, we can assume PCB/CCL will be further upgraded to meet the higher requirement.**

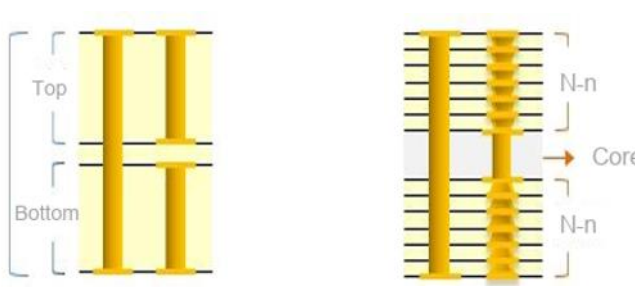
FIGURE 10: PCB SPECIFICATIONS FOR TPU

	ViperFish	GhostFish	SunFish	ZebraFish	HumuFish	TPU v10
CCL material grade	M5-m6	M7	M8 (signal) + M6 (power)	M8 (signal) + M6 (power)	M8/M9	M9/M10
Production schedule for PCB		4Q25	4Q26	1Q27	2028	2029
Layer count	20-22	26	32 (15+2+15)	34 (16+2+16)	~30-40	TBD
Layer count change	-	18%	23%	31%	-	-
	Multi-layer	Multi-layer	N+M	N+M	HDI	HDI
	N+M sequential lamination is inherently more challenging than a standard multilayer build because it requires multiple lamination cycles, yield rates could fall by 15%-30%.				HDI PCB requires fewer layers owing to its high density, but it is more challenging than N+M because it requires multiple cycles of laser drilling, sequential buildup and ultra-fine feature formation. That would see further yield rate loss.	
ASP Index	1	1.3	2.2	2.1	2.5	~5-6

Source: Aletheia/TAG

We see an increasing trend for adopting HDI in advanced AI accelerators: 1) it shortens high-frequency signal paths, reducing signal attenuation and latency, 2) it increases circuit density through microvia technology, including blind and buried vias, 3) it saves internal space and 4) it optimizes heat dissipation efficiency. HDI normally requires fewer layers compared to conventional multi-lam PCB due to its higher density, but HDI PCB is still more challenging because it requires multiple cycles of laser drilling, sequential buildup, and ultra-fine feature formation. The longer processing time and lower yield rate will further drive up content value for PCB.

FIGURE 11: HDI IS MORE ADVANCED THAN MULTI-LAM PCB

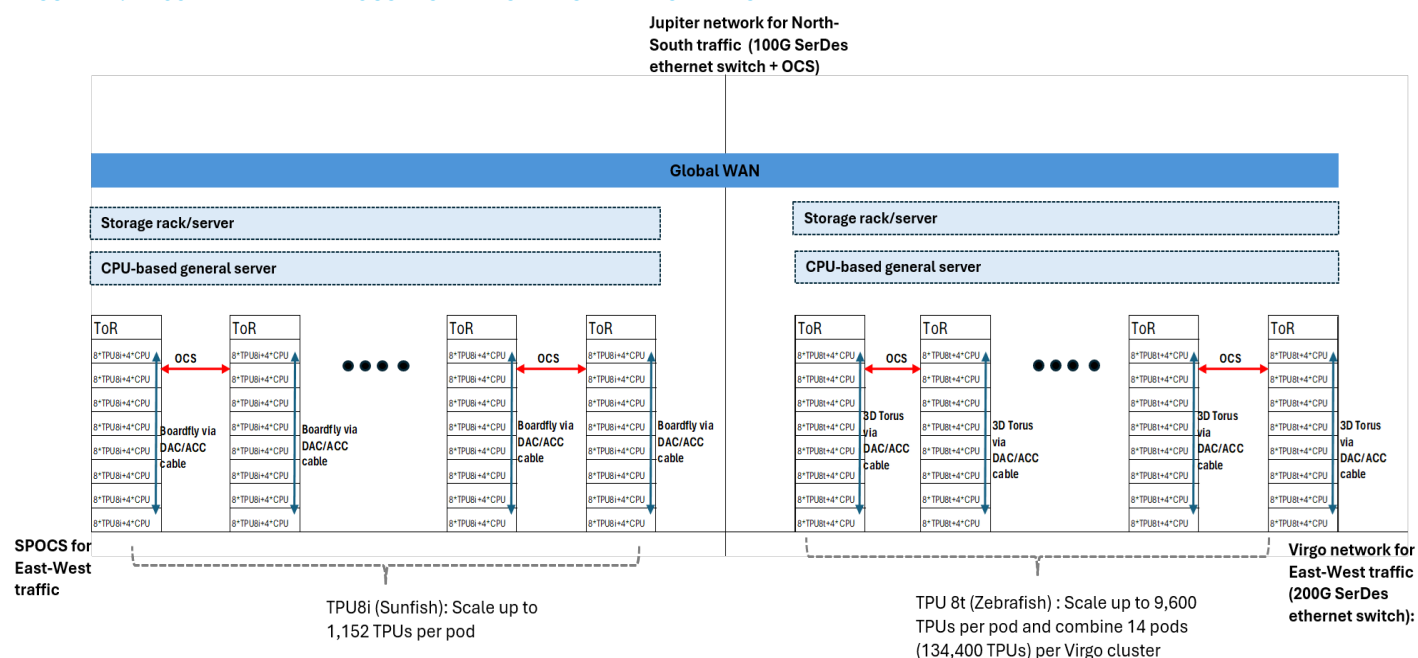
	Multi-lam	HDI
		
Process		
End Product	AI accelerators, switches, networking	for more advanced AI accelerators
Layers	30-50L+	20-40L
Strengths	Optimized for high layer counts (HLC), large panel sizes and power/signal separation (N+M)	Optimized for fine-line routing and high-density interconnects

Source: Aletheia/ TAG

Increasing networking growth will be another pillar of growth

ToR (Top-of-Rack) switch demand for AI server systems is set to grow the fastest during 2026–28 as architectures scale-up and bandwidth requirements rise. As scale-up bandwidth increases, data move outward more quickly, shifting bottlenecks in the scale-out network. Google's TPU v8 network architecture is a good example as illustrated below.

FIGURE 12: MLCC DEMAND IN VARIOUS AI SERVERS AND GENERAL SERVERS



Source: Aletheia/TAG

Adoption of 800 Gbps switches is set to accelerate in 2026–27, with CCL moving from M7 to M8 and PCB architecture shifting from multilayer to N+M. As a result, **we estimate the 800 Gbps switch PCB ASP will rise ~275% to ~USD1,500 from USD350–400 in 400 Gbps systems.** The layer count will increase by 35%–40%, with the remaining value uplift driven by higher-grade materials and more complex structures. For 1.6T switches, PCB ASP is

expected to reach USD3,000–4,000 as the layer count expands to 44–50L and CCL upgrades to M9 (potentially mixed with M8 in power layers).

At the Hot Chips Conference, a premier annual semiconductor industry event, at Stanford University at the end of August, Google further disclosed the networking architecture for its upcoming TPU8t (Zebrafish – MediaTek) and TPU8i (Sunfish – Broadcom). In addition to upgrading its Jupiter network (north-south traffic) architecture, which connects storage, general servers, and its global WAN with higher bandwidth, lower latency, and fewer hops, Google also detailed its new Virgo network (east-west traffic). Through the co-design of network architecture, optical switching technology, and the software orchestration layer, the Virgo network facilitates scaling the maximum TPU cluster size to 1m units (up from 100,000 units in TPU v7 – Ironwood).

This upgrade highly relies on networking hardware advancements, particularly in optical communications. Our deep-dive research into Google’s new TPU network architecture leads to the conclusion that demand will increase for ethernet switches and optical transceivers.

FIGURE 13: OPTICAL NETWORK HARDWARE DEMAND

Component ratio to TPU/Pod	TPU8	TPU7	Remark
Ethernet switch bandwidth -Tb per TPU	2.0	0.4	
Transceiver modules per TPU	3.0	1.8	
Total OCS demand in product lifecycle (Mix of different specs.)	258,542	20,833	We assume lifecycle volume for TPU7 is 4m and TPU8 is 14-15m units.
Optical cable per TPU	18.2	8.0	
DAC/ACC cable per TPU	1.9	1.5	TPU7 uses 100Gbps per lane and TPU8 uses 200Gbps per lane.

Source: Aletheia Capital/TAG

- For switches: TPU8 will start to use 200Gbps SerDes when it was only 100Gbps SerDes in the TPU7 generation. This will require a higher requirement of CCL Df/Dk and PCB layer count and layout. In addition, switch demand will increase to 2Tbps of bandwidth per TPU, rise from 0.4 Tbps in TPU7.
- For transceiver modules: we estimate TPU8 will adopt lots of 800Gbps transceivers and may partially adopt 1.6Tbps transceivers. In addition to that, transceiver module demand per TPU could rise to 3.0-3.5 from 1.8 in TPU7.

FIGURE 14: PCB SPECIFICATIONS FOR SWITCH

	100 Gbps	400 Gbps	800 Gbps	1.6 Tbps
CCL material grade	M6	M7	M8	M8/M9
PCB volume production schedule	-	2022-23	2Q26	2Q27
Layer count	12~18	26~28	36~40	44~50
Layer count change	-	65-75%	35-40%	20-25%
Process highlight	Multi-layer	Multi-layer	N+N	N+M
	N+M sequential lamination is inherently more challenging than a standard multilayer build because it requires multiple lamination cycles. The yield rate could see a 15-30% loss			
PCB ASP (USD)	180	400	1,500-2,000	3,000-4,000
PCB ASP change	-	122%	275%	94%

Source: Aletheia/TAG

Who will benefit most in the Google TPU supply chain?

ISU Petasys is best positioned to capture this growth given its high exposure to networking PCBs (50%–60% of sales) and its role as a key PCB supplier to Google (50%+ of sales). We also consider EMC another great proxy for the TPU supply chain, because of 1) its leading position in high-end CCL, 2) its increasing market share in ASICs (Google, AWS) as well as AMD, and 3) its better margin/profitability along with favorable product mix and price hikes. TTM is another key supplier to Google TPU for N+M products, and we expect its business to accelerate from 2H26. If Google confirms LTAs with its PCB/CCL vendors, we think it will further drive-up the supply chain's P/E with more stable backlogs and longer order visibility.

Price hikes will continue; LTAs are ongoing

Another key growth driver to lift PCB/CCL makers' profitability is price hikes. We have witnessed several rounds of price hikes already this year and expect more. Besides, our channel check suggests that CSPs are working closely with PCB/CCL makers for LTAs to secure advanced capacity and smooth the production process. This section will discuss these two issues and implications for the supply chain.

Price hikes will likely continue

Our supply chain checks suggest that CCL makers are increasing prices by 50%-100% YTD to reflect 1) organic material upgrades under the supply constraint, 2) a shortage in upstream materials supply and price hikes (resin, copper foil and glass cloth), and 3) longer and more complex laminate process for more advanced CCL. We see price hikes across the board from all production lines from M8 material to M4 because tier 1 makers are allocating more capacity for advanced materials (which requires a longer processing time resulting in lower production volume). As such, entry-level products will be excluded and will partially shift to second-tier vendors.

Based on current visibility, we expect more waves of price hikes in CCL in 2H26 and 1H27. We think each round will be 10%-15%. We assume better profitability for CCL makers in future quarters owing to 1) higher UTR under a supply constraint, 2) continued price hikes and 3) favorable product mix along with product migration.

Are LTAs becoming new norm for PCB/CCL?

Given the tight supply conditions and strong PCB/CCL demand for the foreseeable future, we are also hearing that CSPs are aggressively approaching PCB and CCL makers for LTAs to secure capacity. Terms will vary from vendor to vendor. However, those discussions send a clear message that end users are securing high-end PCB/CCL capacity by prepayment and LTAs for the next few years, because the overall industry may continue to remain undersupplied. Also, we reckon prepayment 1) will ease supply chain pressure on capex/cash flow, 2) make the supply chain more willing to invest under longer order visibility and secured shipment volume, and 3) improve supply chain management with better profitability. Below are typical LTAs between memory suppliers and their key customers.

FIGURE 15: DETAILS OF LTAS BETWEEN MEMORY SUPPLIERS AND THEIR CUSTOMERS (AS OF SEP 2026)

	Samsung	SKHY ³	Micron	Kioxia	SNDK
Duration	5 years	5 years	3-5 years ⁴	3+years	4 years
Targeted coverage	60-70%	50-60%	35% ⁵	~50% in 2H26	2/3 in 27/28
Customers	5+5 ¹	10	26	3	8
Ceiling price	Yes	No ³	Yes	No	Yes
Floor price	Yes	Yes	Yes	Yes	Yes
Deposit/ prepayment	Yes ²	No	\$32bn	No	\$16.5bn
Total value	N.A.	N.A.	\$150bn	N.A.	\$94bn

Note: 1- 5 AI customers confirmed as of 2Q26. Five more clients to be confirmed in 3Q26. 2-Cash payment is 25% of value of contract (covering DRAM& NAND). 3- SKHY's LTA doesn't apply to NVDA as the HBM ASP is determined annually and server DRAM (SoCAAM) is semi-annual or quarterly. ASP offered to NVDA will be at a discounted to market price. 4-Auto customers signed 3-year contract while others are on five years term. 5-20% for DRAM and 33% for NAND by 2030.

Source: Aletheia/TAG

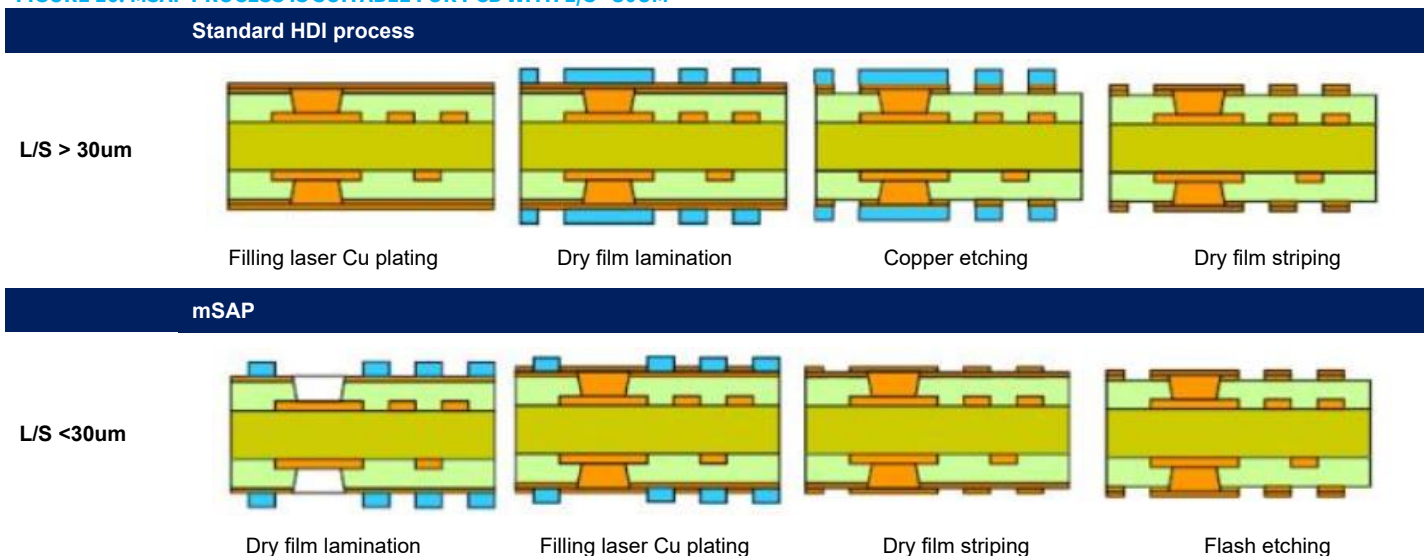
New trends/developments in PCB supply chain and implications

Given the fast development of PCB design, there is lots of noise in the supply chain about the future PCB demand outlook. This section will discuss the latest developments in the PCB industry and their implications for the supply chain, including **mSAP** adoption in optical, pros and cons for **PTFE** material (hybrid/pure) as well as **CPO**'s impact on the CCL supply chain.

mSAP is becoming a standard for optical beyond 1.6T

mSAP (Modified Semi-Additive Process) is not a new technology. It is an advanced PCB manufacturing method that builds fine circuit traces by electroplating copper onto a thin layer rather than etching away thick copper. It allows PCB to further shrink its L/S to below 30um (compared to 30-50um for traditional etching). As such, PCB can support higher density to support advanced electronic devices with a smaller/complex form factor.

FIGURE 16: MSAP PROCESS IS SUITABLE FOR PCB WITH L/S <30UM



Source: Aletheia/TAG

To meet the requirement for higher integration and miniaturization, optical modules are increasingly adopting mSAP to reduce product size and signal loss. Our supply chain check suggests that 1) we are seeing some 800G switch mSAP adoption, but it is more like an optional solution to ally with a specific form factor according to a vendor's requirement, and 2) mSAP is becoming a mandatory process from 1.6T given the L/S is expected to shrink to below 30um.

FIGURE 17: MSAP IS BECOMING A STANDARD FROM 1.6T

	400G	800G	1.6T	3.2T
Layer (HDI)	10L	12-14L	14-18L	20-22L
L/S	40-50um	30-40um	20-30um	10-15um
CCL	M4/M6	M6/M7	M8+	M9+
Suppliers	SCC, Unimicron	SCC, ZDT, Unimicron, Compeq, Kinwong, WUS	SCC, ZDT, Unimicron, Compeq	SCC, ZDT, Unimicron
mSAP adoption	NO	Optional	YES	YES

Source: Aletheia/ TAG

Implications for supply chain: The key bottleneck for mSAP is more in the 1) upstream materials from Japan vendors, including carrier copper foil from Mitsui Kinzoku and Nittobo's low-CTE glass fabric and 2) selected equipment like laser drilling machines and laser direct imaging (LDI) machines. The production lead-time has increased to 10-12 months (vs the normal 4-6 months).

For PCB supply, we believe SCC, ZDT and Unimicron will be the key beneficiaries of mSAP. Compeq is also being qualified for its mSAP solution for 800G/1.6T switches, based on our checks.

FIGURE 18: MAJOR PLAYERS' PROGRESS WITH MSAP

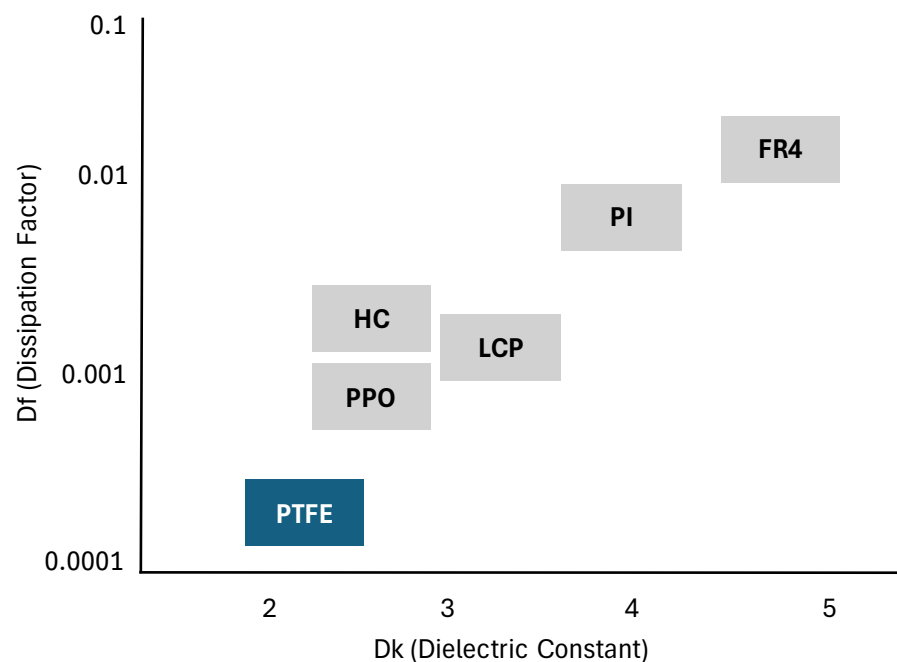
SCC	SCC is a leader in the mSAP process among Chinese PCB makers. The company recently announced a RMB4.37bn private placement to expand AI-related PCB capacity (including mSAP).
ZDT/Avary	ZDT/Avary is aiming to become the world's largest mSAP supplier by expanding capacity 5x in the coming years at all its major production sites (including China, Taiwan and Thailand).
Unimicron	Unimicron raised its 2026 capex from TWD25bn to TWD34bn in its latest earnings call. According to the company, 70% will be used in ABF substrate and 30% for HDI, mSAP and new production site in Thailand.
Compeq	Compeq has passed qualification for mSAP in high-end switch products and started to enter into mass production from 2H26. The company will continue to expand mSAP capacity in both its Taiwan and China (Chongqing) sites.

Source: Aletheia Capital, TAG

Potential PTFE developments in the future

Nvidia is testing hybrid material (PTFE + M9) in the Rubin Ultra generation for switch trays to meet up with higher requirements of data transmission and may consider pure PTFE material for the backplane in the next generation. CCL makers are testing and sampling several solutions at this stage, and we will have better visibility into 1H27.

FIGURE 19: PCB'S MAJOR RESIN MATERIAL DK/DF COMPARISON: PTFE HAS BEST DK/DF



Source: Aletheia/TAG

M9 materials based on PPO/HC resin are no longer sufficient to meet Nvidia's performance requirements. PTFE has better/stable signal performance (low Dk/Df) and the current hybrid solution with no glass cloth can reduce the usage of Q-glass, which is under a supply constraint. However, the hybrid solution is difficult to process (lamine/drill), making its yield rate very low and higher manufacturing costs. Also, the new material will take longer time for qualification.

Implication to supply chain: We don't think PTFE will become mainstream material in AI servers in the near future considering its low yield rate and high cost. It is more likely to be adopted in areas which require extremely low Dk/Df like Nvidia is testing it for switch boards and potentially backplane.

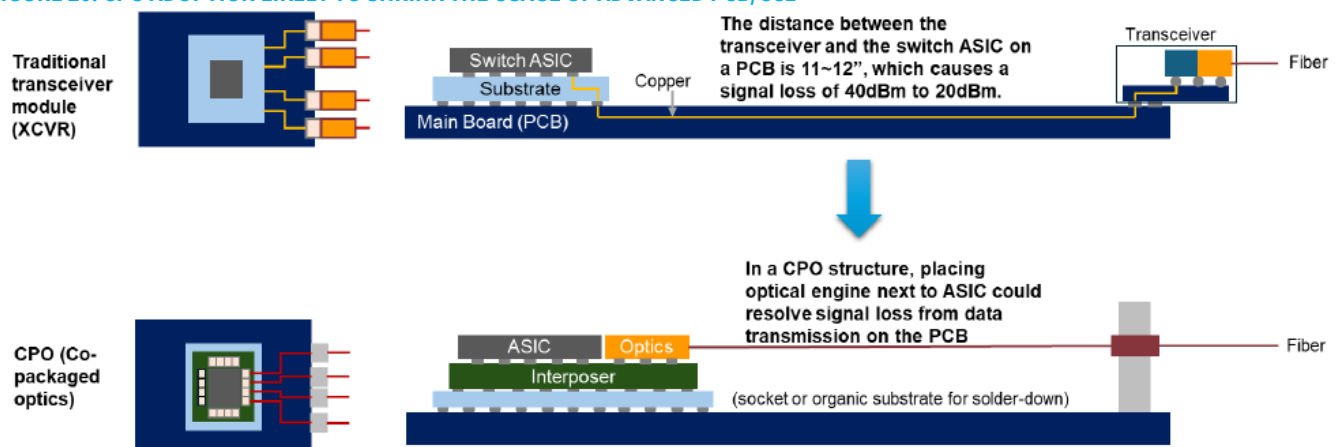
For PTFE CCL supply, we think Shengyi will be the first source given its track record with Nvidia. EMC/Taiflex camp will likely be the second source potentially (EMC supplies M9 PP and Taiflex supplies PTFE CCL). For PCB supply, we think the major suppliers will be Chinese-based PCB vendors (SCC, Kinwong, WUS and VGT for example) and the yield rate is still low (below 30%).

CPO may trigger CCL de-spec?

CPO is an advanced heterogeneous integration of optics and silicon devices in a single form. It enables the integration of both PIC and EIC and packages them in a modular form factor along with switches or XPU (including ASIC) via substrate or interposer to support high-speed transmission and power saving. In traditional systems, electronic signals are converted to optical signals with signal loss and higher power consumption. CPO, however, places the optical engines closer to the switch ASIC or xPU, reducing the distance that signals need to travel and improving the overall performance.

Most common data transmission involves using a transceiver module to input data into the switch ASIC for computing, which must pass through the PCB. For switches, the PCB distance between the transceiver module and the switch ASIC is approximately 11 to 12 inches or $\pm 300\text{mm}$. This results in signal loss and reduces the switch ASIC's signal integrity from 40 dBm to 20 dBm, implying a 50% signal loss. This is despite industry efforts to develop advanced low Df/Dk CCL to reduce signal loss and latency. As such, placing the optical engine next to the switch IC will efficiently resolve this problem as illustrated in figure below.

FIGURE 20: CPO ADOPTION LIKELY TO SHRINK THE USAGE OF ADVANCED PCB/CCL



Source: Aletheia/TAG

Implication for the supply chain: The CPO's purpose is to minimize the data transmission on the PCB, allowing high-speed data transmission to take place within the IC level. In this case, PCB in switch systems may no longer require the low-Df/Dk CCL material compared to its use of high-end low Df/Dk CCL (M6-M8) for 400Gbps and 800Gbps switch today. As such, some people may be concerned that the adoption of CPO will slow down the further migration of CCL.

More impact on CCL side but manageable

As discussed above, we believe the adoption of CPO will impact the CCL side more (material change) instead of the PCB side. However, we don't think it will become an immediate threat and the overall impact should be manageable, because 1) The design of CPO is not finalized and still faces challenges in higher manufacturing costs, lower yield rate, and reliability issues, 2) Nvidia is more aggressive in CPO solutions while other vendors (like Google) still stick with existing solutions, and 3) the overall CCL industry is undersupplied and material upgrades of more and more new applications are taking place in to fulfill future demand.

Hence, our conclusion is 1) CPO penetration takes time and the initial impact will be more in the Nvidia supply chain instead of the entire AI supply chain for CCL makers, and 2) the overall trend for CCL material upgrades will remain strong as other non-CPO application demand remains strong and the overall industry is still undersupplied.

ISU Petasys

#AI #Hardware

↔ Buy
α Portfolio

Price: **KRW 121,200**

Target Price: **KRW 270,000**

Upside: **123%**

Total return: **123%**

Risk: **High**

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ISU Petasys

Bloomberg	007660 KRX
RIC	007660.KRX
Market cap (USDbn)	6.6
Average daily T/O (USD mn)	71

More to Come

We transfer coverage and reiterate BUY for ISU and maintain a TP at KRW270,000 on 30X FY27/28E PER. We remain optimistic for its accelerated revenue and earnings growth into 2028E, driven by increasing PCB design complexity and accelerating deployment of Google's TPU and expanding demand in high-speed networking. We believe the exponential growth compounded by both unit growth and content value upgrades will continue to lead to the growth of its AI PCB business and offset the concerns about market share loss. Not to mention there will be more ASIC projects to come to further boost growth and diversify its concentration. We view ISU as the best proxy to capture solid growth from both Google TPU and networking upgrades thanks to its high exposure, as well as leading high-end PCB technology and strategic geographic presence in Korea.

TPU to lead growth, but more to come

We expect ISU's revenue/earnings to increase sequentially from 2H26 to the entire 2027 along with the Google TPU v8 ramp up. Despite concerns about market share loss, we view that 1) the absolute shipment/revenue contribution from Google will still be more than double, considering the TAM is growing 4-5x with higher unit shipment (3.6m to 8.5/12.0m) and higher content value (80%-150% higher) for both of compute and networking, and 2) less involvement in Google's general server business. The company is also working on customer diversification; MSFT's ASIC will become ISU's next key growth driver from 2028 onward.

New capacity is on the way to support future growth

ISU is expanding capacity aggressively to meet the strong demand: 1) an earlier Fab 6 ramp in 1Q27 for a 25% capacity increase, 2) A 20-25% capacity increase by 2Q28 after Fab 5 expansion for HDI to secure annual sales capacity of W2tn, according to the company. Fab 7 is also under discussion if HDI demand comes in stronger than expected. Its core MLB (multi-lam board) products will remain in strong demand while new HDI products are under development. The company doesn't believe the increasing adoption of HDI will cannibalize demand for MLB. Also, we see continued price hikes (along with upstream material shortage and a strong end demand outlook) and better product mix will drive ISU's profitability to further expand in FY27/28 under a higher UTR.

BUY with unchanged TP of KRW270,000

ISU is uniquely positioned to capture two major trends due to its high exposure to Google and networking. We project its EPS will triple from FY26 to FY28 with larger revenue scale and a favorable product mix. We have a BUY and an unchanged TP of KRW270,000 on 30x FY27/28 P/E. We roll over our valuation period to the average of 2027/28 with longer order visibility and more AI PCB projects to come, while we lower our multiple slightly from 35x to 30x to align with regional peers.

KEY FINANCIAL AND VALUATION MATRIX

Dec 31 (KRW bn)	2022	2023	2024	2025	2026E	2027E	2028E
Revenue	643	675	837	1,088	1,795	3,254	4,567
EBITDA	130	78	125	234	395	734	1,085
Op Profit	117	62	102	205	356	673	987
Net Profit	102	48	74	160	273	537	793
EPS	1,620	755	1,171	2,278	3,766	7,318	10,803
Consensus EPS	-	-	-	-	3,784	5,995	7,631
PER (X)	74.8	160.5	103.5	53.2	32.2	16.6	11.2
PBR (X)	34.5	28.7	23.4	11.3	8.6	5.8	3.9
Dividend Yield (%)	0.1	0.1	0.1	0.2	0.3	0.6	0.9
P/E (%)	57.7	10.5	24.0	20.6	20.8	12.4	11.0

FIGURE 1: SALES ASSUMPTION FOR ISU PETASYS

(KRW bn)	Sales				YoY		
	2025	2026E	2027E	2028E	2026E	2027E	2028E
ISU Petasys (standalone):	930	1,503	2,914	4,169	62%	94%	43%
- Networking	286	458	721	928	60%	57%	29%
- Datacenter	644	1,045	2,193	3,241	62%	110%	48%
ISU Petasys Hunan (CHN)	227	348	388	440	53%	12%	13%
ISU Petasys Corp. (US)	22	33	40	46	52%	21%	15%
Adjustment	-91	-88	-88	-88	-3%	0%	0%
Total ISU sales	1,088	1,795	3,254	4,567	65%	81%	40%
ISU Petasys (standalone):							
- Networking	286	458	721	928	60%	57%	29%
- Datacenter	644	1,045	2,193	3,241	62%	110%	48%
Mix							
- Networking	31%	30%	25%	22%			
- Datacenter	69%	70%	75%	78%			

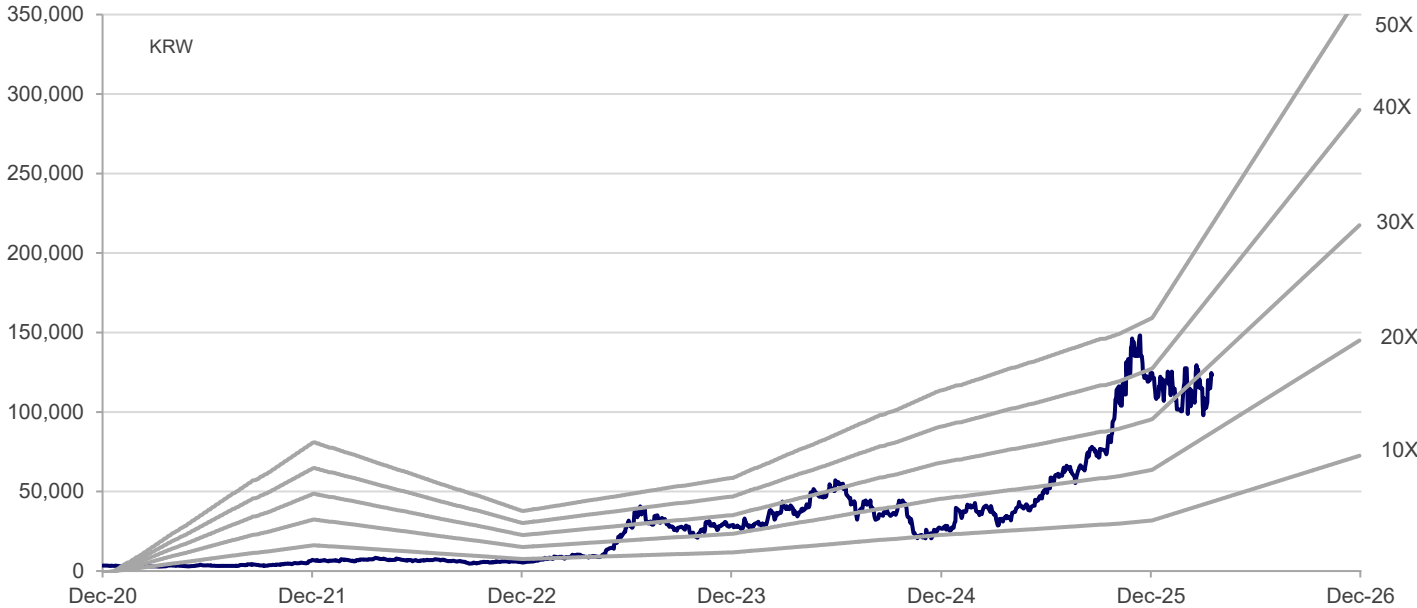
Source: Company, Aletheia/TAG

FIGURE 2: EARNINGS MODEL FOR ISU PETASYS

(KRW bn)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Sales	340	380	455	620	664	739	873	978	837	1,088	1,795	3,254	4,567
Gross profit	84	95	120	165	171	191	227	254	157	282	464	843	1,190
Opex	17	18	28	45	36	39	45	49	55	78	108	169	203
Operating profit	67	77	91	120	135	152	182	205	102	205	356	673	987
Non-OP	-3	0	-3	-1	1	3	5	7	-10	-3	-7	16	30
Pretax profit	64	77	89	120	136	155	187	211	92	202	349	689	1,017
Net profit	50	60	69	93	106	121	145	165	74	160	273	537	793
EPS (KRW)	687	813	964	1,302	1,447	1,642	1,982	2,247	1,171	2,278	3,766	7,318	10,803
Growth													
Sales QoQ	14.2%	11.6%	19.7%	36.5%	7.1%	11.3%	18.0%	12.0%	-	-	-	-	-
Sales YoY	34.8%	57.4%	53.5%	108.2%	95.3%	94.6%	91.9%	57.6%	23.9%	30.0%	65.0%	81.3%	40.4%
EPS QoQ	16.2%	18.3%	18.6%	35.0%	11.2%	13.5%	20.7%	13.4%	-	-	-	-	-
EPS YoY	13.4%	85.6%	49.9%	120.3%	110.6%	101.9%	105.6%	72.6%	55.1%	94.5%	65.3%	94.3%	47.6%
Margin													
Gross margin	24.8%	25.1%	26.3%	26.6%	25.7%	25.8%	26.0%	26.0%	18.7%	25.9%	25.9%	25.9%	26.1%
Opex	5.1%	4.8%	6.2%	7.2%	5.4%	5.3%	5.2%	5.1%	6.5%	7.1%	6.0%	5.2%	4.5%
OP margin	19.8%	20.3%	20.1%	19.4%	20.3%	20.5%	20.8%	20.9%	12.2%	18.8%	19.8%	20.7%	21.6%
Non-OP	-0.8%	-0.1%	-0.6%	-0.1%	0.2%	0.4%	0.6%	0.7%	-1.2%	-0.3%	-0.4%	0.5%	0.7%
Net margin	14.8%	15.7%	15.2%	15.0%	16.0%	16.3%	16.7%	16.9%	8.8%	14.7%	15.2%	16.5%	17.4%

Source: Company, Aletheia Capital, TAG

FIGURE 3: 12-MONTH FORWARD PER BAND FOR ISU PETASYS



Source: Aletheia Capital, TAG

Financial Statements

INCOME STATEMENT							BALANCE SHEET						
Dec 31 (KRW bn)	2023	2024	2025	2026E	2027E	2028E	Dec 31 (KRW bn)	2023	2024	2025	2026E	2027E	2028E
Revenue	675	837	1,088	1,795	3,254	4,567	Cash/ST investment	49	80	164	224	366	412
COGS	-559	-680	-806	-1,331	-2,411	-3,377	Inventory	176	187	299	380	536	675
Gross profit	116	157	282	464	843	1,190	Receivable	152	229	214	359	723	1,015
Operating expense	-54	-55	-78	-108	-169	-203	Current assets	384	523	863	1,150	1,812	2,288
EBITDA	78	125	234	395	734	1,085	Fixed assets	192	202	307	434	620	807
Operating profit	62	102	205	356	673	987	Goodwill & intangibles	4	7	9	12	16	21
Interest expense (income)	-8	-9	-3	-1	24	40	Other LT assets	46	59	110	110	110	110
Other expense (income)	1	-1	0	-6	-9	-10	LT assets	243	267	427	556	746	938
Profit before tax	55	92	202	349	689	1,017	Total assets	626	790	1,290	1,706	2,558	3,227
Tax	-7	-18	-41	-77	-152	-224	Payable	105	167	204	350	689	614
Net profit (parent)	48	74	160	273	537	793	Other liability	185	211	212	177	180	185
FD EPS (KRW)	755	1,171	2,278	3,766	7,318	10,803	Current liabilities	291	377	416	527	869	799
Change							LT liabilities	69	85	117	167	168	167
Revenue	5%	24%	30%	65%	81%	40%	Total liabilities	360	463	533	694	1,036	966
Gross profit	-29%	34%	80%	65%	81%	41%	Common shares	144	144	425	425	425	425
Operating profit	-47%	64%	101%	74%	89%	47%	Retained earnings	114	172	326	581	1,091	1,830
Net profit	-52%	55%	117%	70%	97%	48%	Equity	267	328	756	1,012	1,521	2,261
EPS	-53%	55%	95%	65%	94%	48%	Liability&Equity	626	790	1,290	1,706	2,558	3,227
CASH FLOW							RATIO & PER SHARE DATA						
Dec 31 (KRW bn)	2023	2024	2025	2026E	2027E	2028E	Dec 31 (KRW)	2023	2024	2025	2026E	2027E	2028E
Net profit	48	74	160	273	537	793	Gross margin (%)	17.2	18.7	25.9	25.9	25.9	26.1
Dep. & amortization	15	24	30	45	70	108	EBITDA margin (%)	11.6	14.9	21.5	22.0	22.6	23.7
Chg in WC	0	-29	-182	-111	-173	-501	Op. Profit margin (%)	9.2	12.2	18.8	19.8	20.7	21.6
Others	-10	16	160	0	0	0	Net Profit margin (%)	7.1	8.8	14.7	15.2	16.5	17.4
OPN cash flow	52.5	84.9	168.8	206.4	434.9	399.2	ROE (%)	19.5	24.9	29.6	30.8	42.4	41.9
Capex	-66	-62	-155	-174	-260	-300	ROA (%)	8.1	10.5	15.4	18.2	25.2	27.4
Chg in investment	5	0	-166	0	0	0	Gross gearing (%)	57.4	58.5	41.4	40.7	40.5	29.9
Others	0	0	-25	0	0	0	Net gearing (%)	59.9	45.0	6.1	3.0	-7.7	-7.2
Investing CF	-61.4	-63.0	-346.7	-174.0	-260.0	-300.0	Asset turn (x)	1.2	1.2	1.0	1.2	1.5	1.6
Chg in debt	18	9	-10	45	-5	0	Leverage (x)	2.4	2.4	1.9	1.7	1.7	1.5
Equity raised	0	0	281	0	0	0	Payable days	67	73	84	76	79	70
Dividend paid	-6	-6	-9	-17	-28	-54	Receivable days	81	83	74	58	61	69
Others	-2	0	1	0	0	0	Inventory days	112	97	110	93	69	65
Financing CF	10.3	3.1	262.2	28.1	-32.6	-53.7	EPS	755	1,171	2,278	3,766	7,318	10,803
FX & other adj	1	7	-2	0	0	0	BVPS	4,218	5,179	10,733	14,120	20,724	30,796
Chg in cash	2	32	82	60	142	45	Net cash per share	-2,528	-2,329	-656	-429	1,586	2,206
Beginning cash	46	48	80	163	223	365	SPS	10,678	13,232	15,442	25,052	44,328	62,219
End cash	48.0	80.4	162.8	223.2	365.5	410.9	FCFPS	61	220	-2,050	458	2,117	932
FCF	3.9	13.9	-144.4	32.8	155.4	68.4	DPS	97	146	230	377	732	1,080

Source: Company, Aletheia Capital /TAG

TTM Technologies

#Initiation #AI #Hardware

Buy
Initiation

Price: \$ 131.87

Target Price: \$ 236

Upside: 78.9%

Total return: 78.9%

Risk: High

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TTM Technologies

Bloomberg

RIC

Market cap (USDbn)

Average daily T/O (USD|mn)

TTMI US

TTMI.OQ

13.9

391

A Unique Proxy Standing for USA PCB

We initiate coverage of TTM with a BUY and a TP of USD236 on a 27X FY27/28E average PER based on SOTP. We anticipate TTM earnings will double from FY26E to FY28E, driven by strong AI demand (N+M) and a solid footprint in aerospace & defense (A&D). TTM's data center and networking segment will enjoy 79% CAGR sales in 2026-28 along with TPU v8 ramping up. We believe its leading N+M position and franchise in the US will allow TTM to secure more AI server opportunities (like Meta) in the future. The company is also expanding its capacity aggressively in both Asia and the USA to capture more value in both AI and A&D space.

Leading PCB maker in the US

TTM is one of the world's top 10 PCB makers and No.1 in the USA, as well as a total solutions provider for aerospace & defense (A&D) systems. We think its strong franchise in the USA and global footprint (capacity in North America, China and Southeast Asia) can help the company to seize opportunities in 1) content value upgrades in AI PCBs from Google and other ASICs, 2) increasing spending in the A&D area and its tight relationship with the US government, and 3) organic growth in medical and automotive spaces and continued M&A strategy.

AI revenue to accelerate from 2H26 thanks to TPU, more projects to come

The company reported 90% YoY growth of its data center and networking segment in 2Q26, and raised full-year guidance, suggesting this segment would double in 2026 with strong N+M demand. We project 2H26 AI-related growth will accelerate at 150%-200% YoY along with TPU ramping up to meet company guidance. As one of the major PCB suppliers to Google TPU, we see huge upside in this business: 1) TPU's PCB ASP has lifted 2.5x from TPU V5 to V8 with more complex design and upgrades, and 2) we estimate TPU unit demand will grow from 3.6m in 2026 to 8.5/12.0m in 2027/28. Thanks to its strong capability in high-end PCBs, we believe TTM will penetrate more AI projects in the future (like Meta's ASIC).

EPS to double from FY26-FY28E; initiate with BUY and a USD236 TP

We forecast TTM earnings will double from FY26E to FY28E with better product mix and increasing AI exposure. We expect increasing demand to accelerate capacity additions, supported by TTM's leadership in high-end PCB technology, its long-term relationship with key customers and unique industry position (the largest US-based PCB maker). We initiate with a BUY and a target price of USD236 by using a 27x FY27/28 average PER. We use SOTP to derive our target multiple as we think it will better capture its dual growth drivers (AI and A&D), as its irreplaceability, in our view, make TTM the best proxy in the US to capture both AI server and national defence trends. Our 2026E EPS is largely inline with its guidance and consensus.

KEY FINANCIAL AND VALUATION MATRIX

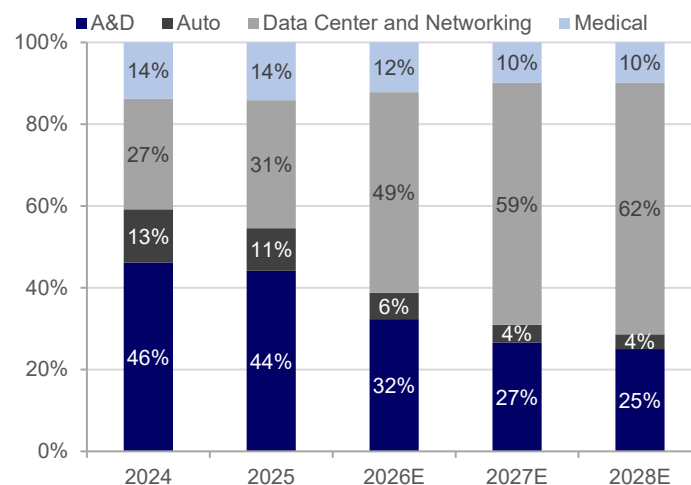
FYE Dec (US\$ mn), non-GAAP	2022	2023	2024	2025	2026E	2027E	2028E
Revenue	2,495	2,233	2,443	2,906	4,423	6,556	7,882
Op Profit	235	199	234	340	664	1,009	1,315
Net Profit	181	140	177	259	505	774	1,025
FD EPS, GAAP	0.9	-0.18	0.54	1.69	3.77	6.45	8.90
FD EPS, non-GAAP	1.8	1.34	1.70	2.46	4.79	7.51	9.97
Consensus EPS, non-GAAP	-	-	-	-	4.84	6.91	7.87
PER (X), non-GAAP	9.0	10.4	14.5	39.9	27.5	17.6	13.2
PBR(X)	1.1	0.9	1.6	5.9	6.5	4.9	3.7
ROA (%)	5.7	4.2	5.2	7.1	12.2	15.9	17.7
ROE (%)	12.2	9.0	11.5	15.6	25.7	31.7	32.2

Largest PCB maker in the USA securing its industry position

TTM is one of the world's top 10 PCB makers and No.1 in the US, as well as a total solutions provider for aerospace & defense (A&D) systems. We think TTM's strong franchise in the US and global footprint (capacity in North America, China and Southeast Asia) can help the company seize opportunities in 1) content value upgrades in AI PCB from Google and other ASICs, 2) increasing spending in the A&D area and its tight relationship with the US government, and 3) organic growth in medical and automotive spaces and continued M&A strategy.

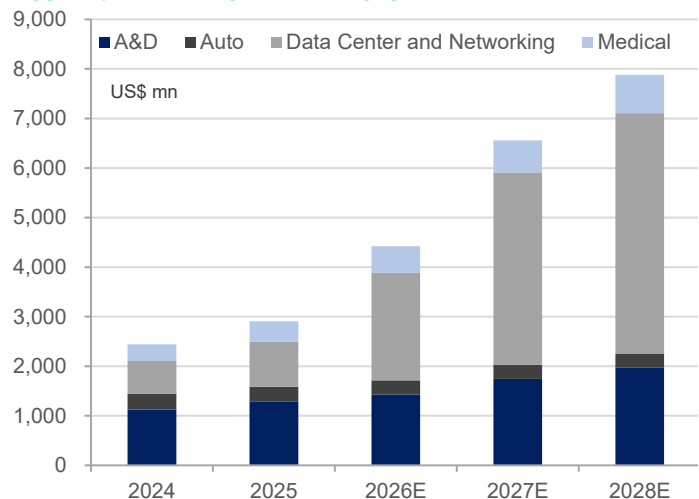
In 2025, 44% of company sales came from from A&D and 56% sales from commercials, which consists of data center and networking (31%), automotive (11%), and medical (14%). This year we started to see exponential growth from high-end AI PCB, thanks to a ramp up of N+M products for AI accelerators. We project data center and networking will become the company's biggest segment from this year, climbing to 49%/59%/62% of total sales in 2026-28E.

FIGURE 1: TTM PRODUCT MIX BY SEGMENT



Source: Company and Aletheia Capital, TAG

FIGURE 2: TTM REVENUE TREND BY SEGMENT



Source: Company and Aletheia Capital, TAG

The company has committed to investing over USD1.2bn from FY25-29 in advanced interconnect/PCB, focusing on 1) N+M products for data center and networking (AI-related applications), 2) ultra-HDI products for the A&D segment, and 3) R&D investments in embedded optics and mSAP, for example. The company is also expanding its capacity aggressively in both Asia and the USA to capture more value in both AI and the A&D space.

FIGURE 3: FUTURE ROADMAP OF TTM PCB PRODUCTS



Source: Aletheia Capital, TAG

Driver 1: advanced AI server PCB from ASIC/GPUs

As one of the major PCB suppliers to Google TPU, we see huge upside for TTM in this business: 1) TPU's PCB ASP has increased 2.5x from TPU V5 to V8 with more complex design and upgrades, and 2) we estimate TPU unit demand will grow from 3.6m in 2026 to 8.5/12.0m in 2027/28. Google's TPU roadmap requires higher ICI bandwidth, HBM density, and compute power, driving a meaningful upgrade on PCB structure and ASP increases. We will elaborate more in the next section.

FIGURE 4: PCB/CCL SPEC AND KEY SUPPLIER FOR AI SERVERS

	Code name	Timeline	PCB specs.	PCB suppliers	CCL specs.	CCL suppliers
Nvidia	GB200	1H25	22L HDI 24L (switch)	VGT, Unimicron VGT, TTM, Unimicron (switch)	M8/M4 M8/M2 (switch)	Doosan Shengyi, EMC (switch)
	GB300	1H26	22L HDI 24L (switch)	VGT, Unimicron VGT, TTM, Unimicron (switch)	M8/M4 M8/M2 (switch) M8/M4	Doosan Shengyi, EMC (switch)
	Vera Rubin	2H26	26L HDI 32L (switch) 44L (mid-plane)	VGT, Unimicron, ZDT VGT, WUS, TTM, Unimicron (switch)	M8/M4 M8 (switch) M8 (mid-plane)	Doosan Shengyi, EMC, Nanya (switch)
	Rubin Ultra	2H27	30L HDI 32L (switch) 44L (mid-plane)	VGT, Unimicron, ZDT VGT, WUS, TTM, Unimicron, SCC (switch)	M8/M6 M8/M9+ PTFE (switch) M9 (mid-plane)	Doosan Shengyi, EMC/Taiflex (switch)
Google	TPU v7 Ironwood	1H26	26L	ISU, LCS, WUS, TTM	M7	Panasonic, EMC
	TPU v8i Sunfish	4Q26	32L M+M	ISU, LCS, WUS, TTM, Dynamic	M8/M6	Panasonic, EMC, Nanya
	TPU v8t Zebrafish	1Q27	34L N+M	ISU, LCS, WUS, TTM, GCE, Dynamic	M8/M6	Panasonic, EMC, Doosan
	TPU v9 Humufish	2028	HDI	ISU, LCS, WUS, TTM, Unimicron, Dynamic	M9/M8	Panasonic, EMC, Doosan
Amazon	Trainium 2e	1H26	26L	GCE, Shengyi, FHT	M8	EMC, TUC
	Trainium 3	2H26	26L/40L 22L (switch)	GCE, Shengyi, FHT, WUS, ZDT	M8	EMC, TUC
	Trainium 4	2H27	34L 30L (switch)	GCE, Shengyi, ZDT	M9?	EMC, TUC, Nanya
AMD	MI450	4Q26	48L	SCC (major), Unimicron, Dynamic	M8	EMC, Doosan
	MI500	4Q27	48-52L	SCC (major), Unimicron, Dynamic	M9	EMC, Doosan
OpenAI	Jalapeno	4Q26	40L	GCE and others	M8?	EMC, Doosan

Source: Aletheia Capital, TAG

FIGURE 5: NVIDIA LPU PCB/CCL SUPPLY CHAIN

	Code name	Timeline	PCB specs.	PCB supplier	CCL specs.	CCL suppliers
Nvidia	LPU	Late 2H26	52L	VGT, WUS, TTM	M9/M8	Shengyi, EMC

Source: Aletheia Capital, TAG

To support the performance upgrades, TPU PCB architecture is shifting from standard multilayer (Ghostfish) to N+M structures using dual-grade CCL and multi-lamination in Zebrafish and Sunfish, and ultimately to HDI PCB in Humufish. These transitions increase PCB ASPs by 30% in Ghostfish, 80%–90% in Sunfish/Zebrafish. Based on our estimates, content value per rack for PCBs is up to USD30k-40k and CCL USD12k-18k for TPU v8 (vs USD20k/7k in v7). We expect another 40%–50% in Humufish with HDI adoption. Into TPU V10, given that TDP will further double to 10kW, we can assume that PCB/CCL will be further upgraded to meet the higher requirement.

FIGURE 6: PCB SPECIFICATION FOR TPU

	ViperFish	GhostFish	SunFish	ZebraFish	HumuFish	TPU v10
CCL material grade	M5-m6	M7	M8 (signal) + M6 (power)	M8 (signal) + M6 (power)	M8/M9	M9/M10
Production schedule for PCB		4Q25	4Q26	1Q27	2028	2029
Layer count	20-22	26	32 (15+2+15)	34 (16+2+16)	~30-40	TBD
Layer count change	-	18%	23%	31%	-	-
	Multi-layer	Multi-layer	N+M	N+M	HDI	HDI
	N+M sequential lamination is inherently more challenging than a standard multilayer build because it requires multiple lamination cycles, yield rates could see 15-30% loss				HDI PCB requires less layers owing to its high density, but it is more challenging than N+M because it requires multiple cycles of laser drilling, sequential buildup and ultra fine feature formation. That would see further yield rate loss.	
ASP Index	1	1.3	2.2	2.1	2.5	~5-6

Source: Aletheia Capital, TAG

We see an increasing trend for adopting HDI in advanced AI accelerators: 1) it shortens high-frequency signal paths, reducing signal attenuation and latency, 2) it increases circuit density through microvia technology, including blind and buried vias, 3) it saves internal space and 4) it optimizes heat dissipation efficiency. HDI normally requires fewer layers compared to conventional multi-lam PCB due to its higher density, but HDI PCB is still more challenging because it requires multiple cycles of laser drilling, sequential buildup, and ultra-fine feature formation. The longer processing and lower yield rate will further drive up content value for PCB.

We believe its leading N+M position and franchise in the US allows TTM secure more AI server opportunities in the future. Apart from Google, we reckon TTM is also one of the key suppliers for Nvidia switch board PCBs and upcoming LPU, which can generate incremental demand for PCB/CCL with its 52L design with M9/M8 materials. Also, our supply chain suggests TTM is qualified to be one of the major PCB suppliers for Meta's ASICs (MTIA series), which will have more meaningful volume in the coming years.

Driver 2: A&D and medical segment growth from both organic and external

Aerospace & Defense represented approximately 46% of 2025 revenue and has become one of TTM's most important structural growth platforms. According to the company, the current program backlog of A&D has reached an historical high level of USD1.7bn. TTM's A&D strategy is to move vertically through the electronics supply chain from PCB/components to module/system solution providers as shown below. This vertical integration is strategically important because TTM can potentially increase its dollar content as customers move from individual components toward increasingly integrated electronic systems.

FIGURE 7: TTM'S VERTICAL INTEGRATION INTO SYSTEM LEVEL



Source: Aletheia Capital, TAG

TTM is investing heavily to expand North American capacity for A&D, including ultra HDI, munitions acceleration, defense growth, PCB reshoring and system design & integration. Given the sensitivity of the business, it is critical to build capacity in the US for core components. Management expects these investments to enable more than USD500mn of incremental projected throughput by 2028.

FIGURE 8: TTM’S POSITION IN A&D AREA

Cross-Domain Integration	Current Positions	Growth Areas	
Space	Missile Warning & Tracking Rad Hard Electronics	Space Payloads & Packaging Rad Tolerant LEO Electronics	↑
Air	Airborne Radar Precision Strike Identification Friend or Foe	Group 1-5 UAS Electronics Next-Gen Radar (AESA)	↑↑
Maritime	Naval Radar & Sensing Shipboard Missiles	Subsea Systems UUV Electronics	↑↑
Land	Air & Missile Defense Tactical Communications Guided Munitions	Counter UAS Systems Edge Compute for Battlefield Networks	↑↑↑

Source: Company and Aletheia Capital, TAG

Apart from organic growth, the company is also using M&A to aggressive expand its A&D business. In August 2026, TTM announced it would acquire Epiq, a solution provider of RF products, space compute, and electronic warfare applications, for approximately USD\$1.1bn (see the [press release](#) for more details). The acquisition will elevate TTM’s vertical integration capabilities for its integrated electronics products and expands its addressable opportunities in high-priority defense mission areas. According to the company, the acquisition is expected to be immediately accretive to EBITDA margins and accretive to non-GAAP diluted EPS during 2028 for the A&D business.

Earlier this year, TTM also announced two acquisitions: Swiss Technology Group (STG) & ILFA. Both are European PCB companies (see the [press release](#) for more details). We view it is a milestone for TTM in establishing its Europe footprint. STG is a Swiss-based PCB company focused on medical (surgical robotics, hearing aids, medical imaging and implantable solutions). Its focus on miniaturized and small form-factor technology is also highly relevant for TTM’s A&D business. ILFA has a 45-year legacy as a German provider of complex PCB solutions spanning A&D, industrial, and medical technology markets.

BUY: a unique proxy standing for USA PCB

We view TTM as a unique proxy among all PCB peers given its presence in the US with leading advanced capacity and tight relationships with key customers. AI PCB will lead the growth at a 75% CAGR in 2026-28 and boost the group sales CAGR at 39%. We forecast TTM earnings will double from FY26E to FY28E with better product mix and increasing AI exposure. We estimate ROE will expand to 26%/32% in 2026/27.

We expect increasing demand to accelerate capacity additions, supported by TTM's leadership in high-end PCB technology, its long-term relationship with key customers and unique industry position (the largest US-based PCB maker). In the meantime, the continued investment in high-end PCB (ultra HDI) and system integration will further secure TTM's leading position in the A&D space as one of the few key suppliers. We initiate with a BUY and a target price of USD236 by using a 27x FY27/28 average PER.

FIGURE 9: OUR ASSUMPTION OF TTM

Sales breakdown (US\$ m)	2024	2025	2026	2027	2028	CAGR 26-28
Aerospace and Defence	1,128	1,284	1,429	1,744	1,970	-
Automotive	318	302	286	286	286	-
Data Center and networking	660	910	2,169	3,880	4,850	-
Medical/Industrial/Instrumentation	336	411	538	646	775	-
Total Sales	2,443	2,906	4,423	6,556	7,882	-
YoY (%)	2024	2025	2026	2027	2028	
Aerospace and Defence	9%	14%	11%	22%	13%	15%
Automotive	-10%	-5%	-5%	0%	0%	-2%
Data Center and networking	35%	38%	138%	79%	25%	75%
Medical/Industrial/Instrumentation	-7%	22%	31%	20%	20%	24%
Total Sales	9%	19%	52%	48%	20%	39%
Mix (%)	2024	2025	2026	2027	2028	
Aerospace and Defence	46%	44%	32%	27%	25%	-
Automotive	13%	10%	6%	4%	4%	-
Data Center and networking	27%	31%	49%	59%	62%	-
Medical/Industrial/Instrumentation	14%	14%	12%	10%	10%	-
Total Sales	100%	100%	100%	100%	100%	-

Source: Aletheia Capital, TAG

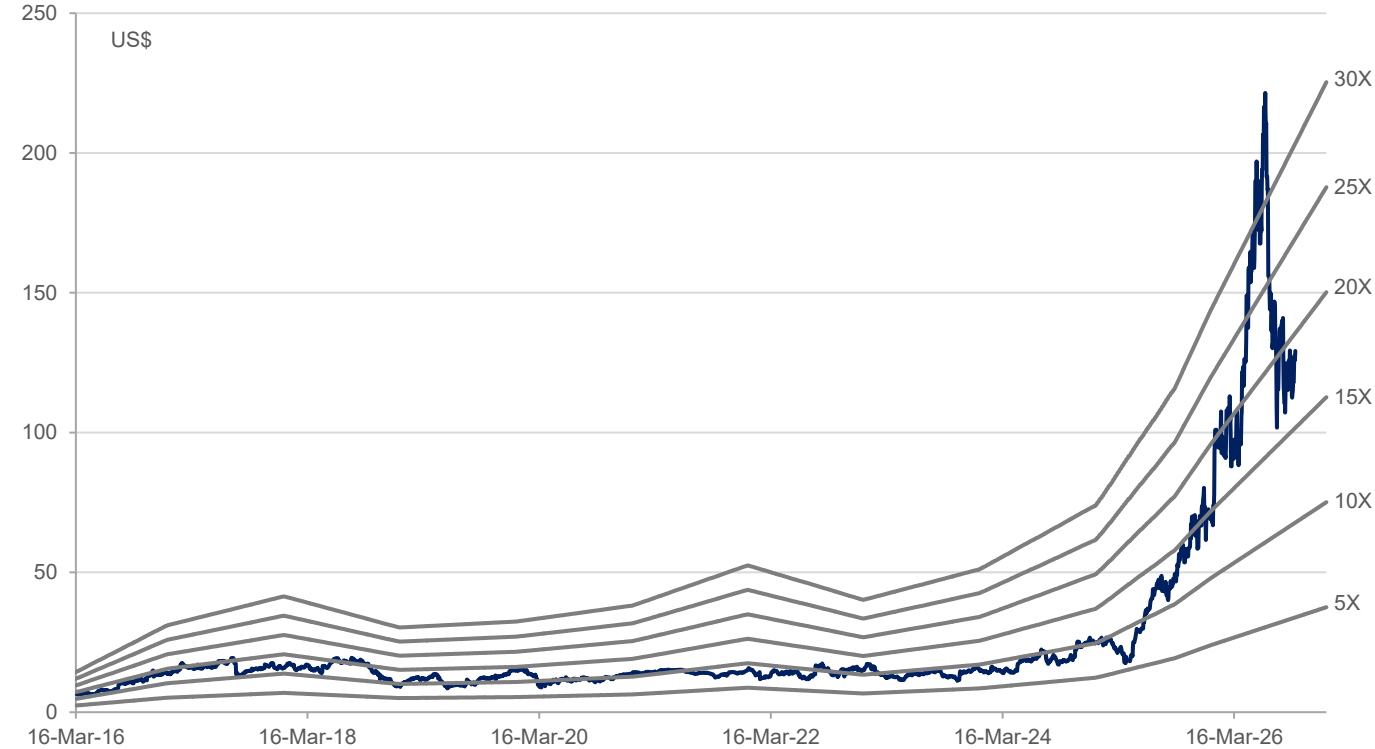
We use SOTP to derive our target multiple as we think it will better capture its dual growth drivers (AI and A&D). We give AI-related business (datacenter and networking) a higher multiple at 30x considering its faster growth rate and higher GPM (25-30%), with A&D and the medical segment at 20x for more moderate growth (low teens) and GPM level (15-25%).

FIGURE 10: SOTP VALUATION

SOTP valuation	2027	2028
Sales mix (%)		
A&D	27%	25%
Auto	4%	4%
Data Center and Networking	59%	62%
Medical and others	10%	10%
G.P. (US\$ m)		
A&D	401	493
Auto	40	40
Data Center and Networking	1,067	1,358
Medical and others	84	101
G.P. mix (%)		
A&D	25%	25%
Auto	3%	2%
Data Center and Networking	67%	68%
Medical and others	5%	5%
Targeted multiple (x)		
Data Center and Networking	30	30
A&D and others	20	20
Implied multiple	27.6	27.8

Source: Aletheia Capital, TAG

FIGURE 11: 12-MONTH FORWARD PER BAND FOR TTM



Source: Aletheia Capital, TAG

Earnings model

FYE Dec (non-GAAP, US\$ mn)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
Aerospace and Defense	338	371	377	343	413	453	460	418	1,031	1,128	1,284	1,429	1,744	1,970
Automotive	68	79	73	66	68	79	73	66	351	318	302	286	286	286
Data Center and networking	305	405	567	892	914	932	964	1,070	490	660	910	2,169	3,880	4,850
Medical/Industrial/Instrumentation	135	148	136	119	162	178	163	143	361	336	411	538	646	775
Others	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Total Sales	846	1,004	1,152	1,420	1,557	1,642	1,659	1,698	2,233	2,443	2,906	4,423	6,556	7,882
Aerospace and Defense	40%	37%	33%	24%	27%	28%	28%	25%	46%	46%	44%	32%	27%	25%
Automotive	8%	8%	6%	5%	4%	5%	4%	4%	16%	13%	10%	6%	4%	4%
Data Center and networking	36%	40%	49%	63%	59%	57%	58%	63%	22%	27%	31%	49%	59%	62%
Medical/Industrial/Instrumentation	16%	15%	12%	8%	10%	11%	10%	8%	16%	14%	14%	12%	10%	10%
Others	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%	0%
Total Sales	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%	100%
Sales	846	1,004	1,152	1,420	1,557	1,642	1,659	1,698	2,233	2,443	2,906	4,423	6,556	7,882
Gross profit	189	220	277	353	377	395	402	418	458	497	618	1,039	1,592	1,992
Opex	81	81	94	119	140	147	147	148	259	263	278	375	583	677
Operating profit	108	138	183	235	236	248	255	270	199	234	340	664	1,009	1,315
Non-OP	-14	-13	-22	-25	-26	-27	-27	-28	-41	-31	-38	-74	-109	-122
Pretax profit	94	126	161	209	210	220	228	242	158	203	303	589	900	1,192
Net profit	80	107	138	180	181	189	196	208	140	177	259	505	774	1,025
Non-GAAP EPS (US\$)	0.75	0.99	1.32	1.73	1.75	1.84	1.90	2.02	1.34	1.70	2.46	4.79	7.51	9.97
Growth	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
Sales QoQ	9.3%	18.7%	14.8%	23.3%	9.6%	5.5%	1.0%	2.3%	-	-	-	-	-	-
Sales YoY	30.4%	37.4%	53.1%	83.4%	84.0%	63.6%	44.0%	19.6%	-10.5%	9.4%	19.0%	52.2%	48.2%	20.2%
EPS QoQ	7.1%	32.0%	33.7%	30.4%	1.5%	4.9%	3.5%	6.2%	-	-	-	-	-	-
EPS YoY	50.0%	70.7%	94.7%	146.6%	133.5%	85.6%	43.6%	17.0%	-23.4%	26.9%	44.7%	94.7%	56.8%	32.7%
Margin	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2023	2024	2025	2026E	2027E	2028E
Gross margin	22.3%	21.9%	24.0%	24.9%	24.2%	24.0%	24.2%	24.6%	20.5%	20.4%	21.3%	23.5%	24.3%	25.3%
Opex	9.5%	8.1%	8.2%	8.3%	9.0%	9.0%	8.8%	8.7%	11.6%	10.8%	9.6%	8.5%	8.9%	8.6%
OP margin	12.8%	13.8%	15.9%	16.5%	15.2%	15.1%	15.4%	15.9%	8.9%	9.6%	11.7%	15.0%	15.4%	16.7%
Non-OP	-1.7%	-1.3%	-1.9%	-1.8%	-1.7%	-1.7%	-1.7%	-1.6%	-1.8%	-1.3%	-1.3%	-1.7%	-1.7%	-1.6%
Net margin	9.5%	10.6%	12.0%	12.7%	11.6%	11.5%	11.8%	12.3%	6.3%	7.3%	8.9%	11.4%	11.8%	13.0%

Source: Company, Aletheia Capital/TAG

Financial Statements

INCOME STATEMENT							BALANCE SHEET						
Dec 31 (US\$ mn), non-GAAP	2023	2024	2025	2026E	2027E	2028E	Dec 31 (US\$ mn)	2023	2024	2025	2026E	2027E	2028E
Revenue	2,233	2,443	2,906	4,423	6,556	7,882	Cash/ST investment	450	504	501	474	516	928
COGS	-1,775	-1,946	-2,288	-3,384	-4,964	-5,891	Inventory	213	225	250	379	574	697
Gross profit	458	497	618	1,039	1,592	1,992	Receivable	414	449	564	867	1,311	1,576
Operating expense	-259	-263	-278	-375	-583	-677	Current assets	1,430	1,607	1,855	2,261	2,942	3,742
EBITDA	361	385	504	845	1,214	1,559	Fixed assets	894	948	1,092	1,312	1,545	1,737
Operating profit	199	234	340	664	1,009	1,315	Goodwill & intangibles	939	862	825	788	751	714
Interest expense	-42	-32	-54	-89	-109	-122	Other LT assets	61	56	68	68	68	68
Other expense (income)	1	1	17	15	0	0	LT assets	1,894	1,866	1,985	2,169	2,364	2,520
Profit before tax	158	203	303	589	900	1,192	Total assets	3,324	3,472	3,840	4,429	5,306	6,261
Tax	19	25	44	84	126	167	Payable	335	406	544	759	999	1,077
Net profit (parent)	140	177	259	505	774	1,025	Other liability	369	403	419	419	419	419
EPS \$, GAAP	-0.18	0.54	1.69	3.77	6.45	8.90	Current liabilities	704	809	962	1,178	1,418	1,495
EPS \$, non-GAAP	1.34	1.70	2.46	4.79	7.51	9.97	LT liabilities	1,109	1,100	1,116	1,123	1,135	1,147
Change							Total liabilities	1,813	1,909	2,078	2,301	2,553	2,642
Revenue	-10.5%	9.4%	19.0%	52.2%	48.2%	20.2%	Common shares	881	911	952	922	882	832
Gross profit	-2.9%	8.5%	24.3%	68.0%	53.3%	25.1%	Retained earnings	782	838	1,016	1,412	2,077	2,993
Operating profit	-15.2%	17.3%	45.6%	95.0%	52.1%	30.2%	Equity	1,511	1,564	1,762	2,129	2,753	3,619
Net profit	-23.0%	27.2%	45.9%	95.1%	53.3%	32.4%	Liability&Equity	3,324	3,472	3,840	4,429	5,306	6,261
EPS, non-GAAP	-23.4%	26.9%	44.7%	94.7%	56.8%	32.7%							
CASH FLOW							RATIO & PER SHARE DATA						
Dec 31 (US\$ mn)	2023	2024	2025	2026E	2027E	2028E	Dec 31 (US\$ mn)	2023	2024	2025	2026E	2027E	2028E
Net profit	-19	56	177	396	665	916	Gross margin (%)	18.5	19.5	20.7	22.8	23.8	24.9
Dep. & amortization	161	150	147	166	204	244	EBITDA margin (%)	9.1	10.9	14.2	16.0	16.7	18.3
Chg in WC	100	-19	-98	-217	-399	-310	Op. Profit margin (%)	1.9	4.8	9.1	12.2	13.6	15.2
Others	-55	49	66	0	0	0	Net Profit margin (%)	-0.8	2.3	6.1	9.0	10.1	11.6
OPN cash flow	187	237	292	345	471	850	ROE (%)	-1.2	3.7	10.7	20.4	27.2	28.7
Capex	-160	-186	-293	-350	-400	-400	ROA (%)	-0.6	1.7	4.9	9.6	13.7	15.8
Chg in investment	6	0	0	0	0	0	Gross gearing (%)	54.5	55.0	54.1	51.9	48.1	42.2
Others	62	40	19	7	12	12	Net gearing (%)	36.1	31.2	28.3	24.7	17.6	2.0
Investing CF	-92	-146	-274	-343	-388	-388	Asset turn (x)	0.7	0.7	0.8	1.1	1.3	1.4
Chg in debt	-7	-1	-4	0	0	0	Leverage (x)	2.2	2.2	2.2	2.1	2.0	1.8
Equity raised / Repurchased	-24	-34	-18	-30	-40	-50	Payable days	70	69	75	70	64	64
Dividend paid	0	0	0	0	0	0	Receivable days	72	64	64	59	61	67
Others	-16	-1	1	0	0	0	Inventory days	38	41	38	34	35	39
Financing CF	-48	-37	-21	-30	-40	-50	EPS	-0.2	0.5	1.7	3.8	6.4	8.9
FX & other adj	0	0	0	0	0	0	BVPS	14.7	15.0	16.7	20.4	26.7	35.2
Chg in cash	47	54	-2	-28	43	412	Net cash per share	-5.3	-4.7	-4.7	-5.0	-4.7	-0.7
Beginning cash	403	450	504	501	474	516	SPS	21.7	23.5	27.6	42.4	63.6	76.6
End cash	450	504	501	474	516	928	FCFPS	-25.0	0.2	-0.2	0.7	1.6	5.4
FCF	-2,566	24	-20	74	163	554	DPS	-	-	-	-	-	-

Source: Company, Aletheia Capital /TAG

Elite Material

#AdvancedMaterial

↔ Buy

Price: **TWD 5,185**

Target Price: **TWD 10,000**

Upside: **93%**

Total return: **93%**

Risk: **High**

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Elite Material

Bloomberg

2383 TT

RIC

2383.TW

Market cap (USDbn)

58.2

Average daily T/O (USD|mn)

362

Leader in CCL

We transfer coverage and reiterate BUY and set a NT\$10,000 target price for EMC, based on 30x FY27/28E P/E, supported by continued material upgrades and price hikes for high-end CCL. EMC is the best proxy to seize strong upside from ASIC AI into production ramping up (exposure to both Google TPU and AWS Trn3) into 2H26/2027. We estimate TPU unit demand will grow from 3.6m in 2026 to 8.5/12.0m in 2027/2028. EMC's capacity will double by the end of 2028 (vs '25) to support a strong demand pipeline and new AI CCL opportunities (more ASIC/GPU projects, HDI adoption and PTFE, for example) We think EMC's leading position will be more secured when migrating to more advanced CCL materials with fewer qualified suppliers.

Unstoppable upgrade trend in CCL

We believe the CCL upgrade cycle will continue to surprise the market because 1) stronger AI device unit growth (especially Google TPU we project 8.5/12m units in 2027/28), 2) significant content value growth via CCL material upgrades (M8/M9, PTFE), 3) wider AI end devices (GPU, ASIC, switches, CPU and LPU, for example) will all require high end PCB/CCL to support performance and 4) better cash flow and profitability via prepayment and LTAs. The overall industry is under a severe supply shortage at this stage, especially high-end products, with the key bottleneck in upstream material (CCL, glass fibre and copper foil) to trigger further rounds of price hikes. We believe EMC's leading position will become more solid when migrating to more advanced CCL materials (M8/M9) with 1) fewer and less qualified suppliers, and 2) additional capacity ready.

Capacity to double by end of 2028 (vs 2025) to support strong growth

EMC's capacity will double by the end of 2028 (vs 2025) to support a strong demand pipeline and new AI CCL opportunities, including 1) further CCL upgrades in TPU v9/Trn 4, 2) increasing adoption of HDI in AI accelerators to benefit EMC, 3) more ASIC/GPU projects like AMD/Meta, for example. M8 is becoming a standard material for AI servers/networking and we will see more advanced materials into 2027/28 (M9/PTFE). CCL is the most critical component in PCB and every generational upgrade implies the ASP will nearly double.

Buy with new TP at NT\$10,000

We see a clear trend of CCL upgrades continuing into 2028 based on current design roadmaps and product pipelines. Its industry-leading position in high-end CCL and new capacity means EMC is the best proxy for upside from ASIC AI into production ramping up. We reiterate BUY and set a new TP at NT\$10,000 with 30x FY27/28 PE. We roll over our valuation period to the average of 2027/28 with longer order visibility and more AI CCL upside for upgrades.

KEY FINANCIAL AND VALUATION MATRIX

Dec 31 (TWD mn)	2022	2023	2024	2025	2026E	2027E	2028E
Revenue	38,673	41,296	64,377	94,261	209,965	365,489	525,236
Op Profit	6,226	7,346	12,152	19,108	59,500	120,734	185,447
Net Profit	5,073	5,488	9,578	14,649	46,159	94,281	144,819
EPS	14.77	15.78	27.32	40.81	128.72	262.85	403.76
Consensus EPS	-	-	-	-	115.10	227.24	353.24
PER (X)	351.0	328.6	189.8	127.1	40.3	19.7	12.8
PB (X)	80.7	67.3	51.8	36.9	21.2	12.1	7.7
Dividend Yield (%)	0.2	0.2	0.3	0.5	1.5	3.0	4.7
ROE (%)	24.3	22.5	30.9	34.2	66.9	78.0	73.0

FIGURE 1: INCOME STATEMENT FOR EMC

(NT\$m)	1Q26	2Q26	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E
Sales	33,067	47,275	58,617	71,005	75,861	83,600	96,958	109,070	64,377	94,261	209,965	365,489	525,236
Gross profit	9,729	16,003	21,340	26,620	28,552	31,871	37,448	42,695	17,970	28,120	73,692	140,565	207,215
Opex	2,601	3,269	3,866	4,456	4,518	4,711	5,154	5,449	5,818	9,012	14,192	19,832	21,768
Operating profit	7,128	12,734	17,474	22,164	24,034	27,160	32,294	37,246	12,152	19,108	59,500	120,734	185,447
Non-OP	66	52	11	20	25	30	35	40	-19	-232	148	130	210
Pretax profit	7,194	12,785	17,485	22,184	24,059	27,190	32,329	37,286	12,133	18,876	59,649	120,864	185,657
Tax expense	1,854	2,912	3,845	4,879	5,291	5,980	7,111	8,201	2,554	4,227	13,490	26,583	40,837
Net profit	5,340	9,873	13,640	17,305	18,768	21,210	25,218	29,085	9,578	14,649	46,159	94,281	144,819
EPS (NT\$)	14.89	27.55	38.03	48.25	52.32	59.13	70.31	81.09	27.32	40.81	128.72	262.85	403.76
Growth													
Sales QoQ	32.7%	43.0%	24.0%	21.1%	6.8%	10.2%	16.0%	12.5%	-	-	-	-	-
Sales YoY	52.5%	110.0%	133.1%	184.8%	129.4%	76.8%	65.4%	53.6%	55.9%	46.4%	122.7%	74.1%	43.7%
EPS QoQ	42.9%	84.9%	38.2%	26.9%	8.5%	13.0%	18.9%	15.3%	-	-	-	-	-
EPS YoY	53.9%	183.9%	244.0%	363.1%	251.5%	114.8%	84.9%	68.1%	73.1%	49.4%	215.4%	104.2%	53.6%
Margin													
Gross margin	29.4%	33.9%	36.4%	37.5%	37.6%	38.1%	38.6%	39.1%	27.9%	29.8%	35.1%	38.5%	39.5%
Opex	7.9%	6.9%	6.6%	6.3%	6.0%	5.6%	5.3%	5.0%	9.0%	9.6%	6.8%	5.4%	4.1%
OP margin	21.6%	26.9%	29.8%	31.2%	31.7%	32.5%	33.3%	34.1%	18.9%	20.3%	28.3%	33.0%	35.3%
Non-OP	0.2%	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	-0.2%	0.1%	0.0%	0.0%
Tax rate	25.8%	22.8%	22.0%	22.0%	22.0%	22.0%	22.0%	22.0%	21.1%	22.4%	22.6%	22.0%	22.0%
Net margin	16.1%	20.9%	23.3%	24.4%	24.7%	25.4%	26.0%	26.7%	14.9%	15.5%	22.0%	25.8%	27.6%

Source: Company data, Aletheia/TAG

FIGURE 2: EMC CAPACITY EXPANSION PLAN

(k sheets/month)	2025	2026	2027	2028
China	4,650	4,650	7,050	8,850
Taiwan	600	900	1,500	1,500
Malaysia	600	600	750	750
Total	5,850	6,150	9,300	11,100
YoY	34%	5%	51%	19%

Source: Company, Aletheia/TAG

FIGURE 3: ASSUMPTIONS FOR EMC SALES AND GM

	2023	2024	2025	2026	2027	2028
Sales mix (NT\$m)						
Infrastructure	20,784	38,292	64,154	166,472	308,975	459,061
<i>Server</i>	9,198	17,703	37,380	111,715	228,836	360,965
<i>Switch</i>	11,014	17,521	20,533	44,610	67,260	82,441
<i>Basestation</i>	572	3,069	6,241	10,148	12,879	15,655
Mobile device	13,745	16,685	20,525	30,884	42,121	51,199
Auto and Industry	6,767	9,400	9,582	12,608	14,392	14,977
Total	41,296	64,377	94,261	209,965	365,489	525,236
Sales mix						
Infrastructure	50%	59%	68%	79%	85%	87%
<i>Server</i>	44%	46%	58%	67%	74%	79%
<i>Switch</i>	53%	46%	32%	27%	22%	18%
<i>Basestation</i>	3%	8%	10%	6%	4%	3%
Mobile device	33%	26%	22%	15%	12%	10%
Auto and Industry	16%	15%	10%	6%	4%	3%
Total	100%	100%	100%	100%	100%	100%
Growth, YoY						
Infrastructure	32%	84%	68%	159%	86%	49%
<i>Server</i>	41%	92%	111%	199%	105%	58%
<i>Switch</i>	24%	59%	17%	117%	51%	23%
<i>Basestation</i>	82%	437%	103%	63%	27%	22%
Mobile device	-7%	21%	23%	50%	36%	22%
Auto and Industry	-17%	39%	2%	32%	14%	4%
Total	7%	56%	46%	123%	74%	44%
Gross margin						
Infrastructure	29.4%	30.2%	31.9%	38.1%	40.6%	41.3%
<i>Server</i>	29.5%	31.4%	33.7%	39.2%	41.6%	42.0%
<i>Switch</i>	29.5%	30.3%	31.3%	38.4%	40.6%	41.4%
<i>Basestation</i>	25.0%	21.9%	23.0%	25.0%	25.0%	25.0%
Mobile device	25.5%	24.3%	26.6%	26.2%	27.0%	27.0%
Auto and Industry	23.5%	24.5%	24.7%	25.6%	25.0%	25.0%
Total	27.4%	27.9%	29.8%	35.1%	38.5%	39.5%

Source: Company, Aletheia/TAG

Financial Statements

INCOME STATEMENT							BALANCE SHEET						
Dec 31 (TWD mn)	2023	2024	2025	2026E	2027E	2028E	Dec 31 (TWD mn)	2023	2024	2025	2026E	2027E	2028E
Revenue	41,296	64,377	94,261	209,965	365,489	525,236	Cash/ST investment	9,259	14,989	20,008	20,556	38,785	78,897
COGS	-29,964	-46,407	-66,141	-136,272	-224,924	-318,021	Inventory	6,135	9,437	16,752	27,254	44,985	63,604
Gross profit	11,333	17,970	28,120	73,692	140,565	207,215	Receivable	17,196	25,865	36,065	69,988	121,830	175,079
Operating expense	-3,987	-5,818	-9,012	-14,192	-19,832	-21,768	Current assets	33,675	51,494	73,980	118,954	206,754	318,735
EBITDA	8,920	14,218	20,942	62,553	124,596	190,176	Fixed assets	16,655	21,387	30,864	48,293	59,820	70,480
Operating profit	7,346	12,152	19,108	59,500	120,734	185,447	Goodwill & intangibles	2,222	2,720	2,495	2,442	2,382	2,322
Interest expense (income)	-198	-315	-216	-281	-199	-119	Other LT assets	875	479	574	1,215	1,205	1,195
Other expense (income)	272	297	-16	429	329	329	LT assets	19,752	24,586	33,933	51,951	63,407	73,997
Profit before tax	7,420	12,133	18,876	59,649	120,864	185,657	Total assets	53,427	76,080	107,913	170,904	270,162	392,732
Tax	-1,931	-2,564	-4,231	-13,496	-26,590	-40,844	Payable	10,489	15,963	24,518	50,514	83,376	117,886
Net profit (parent)	5,488	9,578	14,649	46,159	94,281	144,819	Other liability	12,584	14,219	21,713	21,623	21,533	21,443
EPS NT\$	15.8	27.3	40.8	128.7	262.9	403.8	Current liabilities	23,073	30,182	46,231	72,138	104,910	139,330
Change							LT liabilities	3,544	10,804	11,247	11,147	11,047	10,947
Revenue	6.8%	55.9%	46.4%	122.7%	74.1%	43.7%	Total liabilities	26,617	40,986	57,477	83,284	115,956	150,276
Gross profit	16.7%	58.6%	56.5%	162.1%	90.7%	47.4%	Common shares	3,432	3,466	3,583	3,583	3,583	3,583
Operating profit	18.0%	65.4%	57.2%	211.4%	102.9%	53.6%	Retained earnings	19,875	25,997	34,744	71,929	138,515	226,766
Net profit	8.2%	74.5%	52.9%	215.1%	104.3%	53.6%	Equity	26,809	35,094	50,435	87,620	154,205	242,456
EPS	6.8%	73.1%	49.4%	215.4%	104.2%	53.6%	Liability&Equity	53,427	76,080	107,913	170,904	270,162	392,732
CASH FLOW							RATIO & PER SHARE DATA						
Dec 31 (TWD mn)	2023	2024	2025	2026E	2027E	2028E	Dec 31 (TWD mn)	2023	2024	2025	2026E	2027E	2028E
Net profit	5,488	9,569	14,645	46,159	94,281	144,819	Gross margin (%)	27.4	27.9	29.8	35.1	38.5	39.5
Dep. & amortization	1,302	1,769	1,849	2,623	3,534	4,400	EBITDA margin (%)	21.6	22.1	22.2	29.8	34.1	36.2
Chg in WC	-2,224	-6,204	-4,901	-18,429	-36,710	-37,359	Op. Profit margin (%)	17.8	18.9	20.3	28.3	33.0	35.3
Others	-1,677	2,129	374	-641	10	10	Net Profit margin (%)	13.3	14.9	15.5	22.0	25.8	27.6
OPN cash flow	2,890	7,263	11,968	29,712	61,114	111,871	ROE (%)	22.5	30.9	34.2	66.9	78.0	73.0
Capex	-3,281	-5,867	-9,490	-20,000	-15,000	-15,000	ROA (%)	11.3	14.8	15.9	33.1	42.7	43.7
Chg in investment	0	0	0	0	0	0	Gross gearing (%)	49.8	53.9	53.3	48.7	42.9	38.3
Others	-776	-34	-398	0	0	0	Net gearing (%)	(2.2)	(0.1)	(5.0)	(3.7)	(14.1)	(25.6)
Investing CF	-4,057	-5,901	-9,888	-20,000	-15,000	-15,000	Asset turn (x)	0.9	1.0	1.0	1.5	1.7	1.6
Chg in debt	2,954	-614	-2,682	-190	-190	-190	Leverage (x)	2.0	2.1	2.2	2.0	1.8	1.7
Equity raised	103	35	117	0	0	0	Payable days	104	104	112	100	109	115
Dividend paid	-2,830	-3,439	-5,894	-8,974	-27,695	-56,568	Receivable days	127	122	120	92	96	103
Others	-118	8,134	11,235	0	0	0	Inventory days	63	61	72	59	59	62
Financing CF	109	4,116	2,776	-9,164	-27,885	-56,758	EPS	15.8	27.3	40.8	128.7	262.9	403.8
FX & other adj	-126	251	163	0	0	0	BVPS	77.1	100.1	140.6	244.3	429.9	676.0
Chg in cash	-1,185	5,729	5,020	548	18,229	40,113	Net cash per share	1.7	0.1	7.0	9.1	60.4	172.8
Beginning cash	10,444	9,259	14,988	20,008	20,556	38,785	SPS	118.7	183.6	262.8	585.4	1,019.0	1,464.4
End cash	9,259	14,988	20,008	20,556	38,785	78,897	FCFPS	4.1	-1.4	6.3	29.5	129.0	270.3
FCF	1,432	-484	2,272	10,564	46,253	96,947	DPS	10.0	16.6	25.0	77.2	157.7	242.3

Source: Company, Aletheia Capital /TAG

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